

Offline and Preference-Based Reinforcement Learning for Robotics: A Survey

Robotics is a natural setting for offline and preference-based reinforcement learning because real-world robot interaction is expensive, slow, and safety-critical, while human judgments often specify task success better than hand-coded rewards. Offline reinforcement learning learns policies from fixed datasets of demonstrations, logs, or prior robot experience; preference-based reinforcement learning learns reward or policy updates from rankings, comparisons, corrections, or other forms of human feedback. This survey synthesizes an arXiv-derived corpus of 300 papers collected during this run and organizes the literature around offline RL objectives, preference and reward learning, imitation from demonstrations, robot manipulation and control, offline-to-online adaptation, safety, benchmarks, and foundation-model interfaces. The survey argues that progress depends on treating data, rewards, uncertainty, and human feedback as one integrated robot-learning pipeline rather than as separate algorithmic components.

Keywords:

offline reinforcement learning; preference-based reinforcement learning; robotics; robot learning; human feedback; imitation learning; safety; benchmarks

1 Introduction

Robots learn under constraints that make online trial-and-error difficult. A robot arm can damage objects, a mobile robot can collide with people, and a dexterous hand can require thousands of expensive resets. Offline reinforcement learning addresses this bottleneck by learning from previously collected data, while preference-based reinforcement learning addresses the reward-design bottleneck by using human comparisons, rankings, corrections, or feedback to specify what successful behavior should look like [5, 14, 131, 138, 165, 168, 204, 236]. Together, these approaches promise scalable robot learning from data and judgment rather than from unrestricted environment interaction.

The two paradigms are complementary. Offline RL provides a way to extract policies from demonstrations, teleoperation logs, autonomous rollouts, simulation data, or mixed-quality datasets. Preference-based RL provides a way to improve reward specification when sparse success signals, hand-engineered costs, or simulator rewards do not match human

intent. Robotics brings additional structure: policies must satisfy physical constraints, recover from distribution shift, handle partial observability, and remain safe during offline-to-online fine-tuning [131, 147, 168, 196, 242, 262, 294, 299].

This survey reviews offline and preference-based reinforcement learning for robotics. We focus on how static data, learned rewards, demonstrations, conservative objectives, uncertainty, and human-in-the-loop updates interact in robot manipulation, locomotion, navigation, and language-conditioned control.

2 Background

2.1 Offline Reinforcement Learning

Offline RL, also called batch RL, learns a policy from a fixed dataset without further environment interaction during training. The central challenge is extrapolation error: a learned value function can assign high value to actions that are poorly supported by data. Conservative objectives, behavior regularization, advantage-weighted regression, model-based

rollouts, and sequence modeling all address this issue in different ways. In robotics, the dataset may contain expert demonstrations, suboptimal play, scripted controllers, teleoperation, simulation, or mixed logs [131, 147, 168, 196, 242, 262, 294, 299].

2.2 Preference-Based Reinforcement Learning

Preference-based RL replaces or augments scalar rewards with human judgments. A user may compare trajectory clips, rank outcomes, label corrections, accept or reject behavior, or provide language feedback. A reward model or policy optimizer then uses these signals to improve control. Preference feedback is attractive for robotics because many tasks are easy to judge but hard to formalize: grasp quality, natural motion, helpful assistance, and social acceptability often require human evaluation [5, 14, 111, 168, 204, 218, 236, 291].

2.3 Robotics Constraints

Robot learning differs from simulated control because data collection has cost, risk, latency, and hardware variation. Policies must generalize across objects, scenes, dynamics, cameras, and embodiments. Offline and preference-based methods therefore need uncertainty estimates, conservative updates, reliable evaluation, and mechanisms for transferring from simulation or logged data to real deployment [111, 131, 165, 168, 204, 218, 236, 291].

3 Methods

3.1 Search Protocol and Corpus Construction

The corpus for this survey was generated during this run from the arXiv structured API. Queries combined the task topic with offline reinforcement learning, batch RL, conservative Q-learning, implicit Q-learning, decision transformers, offline-to-online RL, safe offline RL, preference-based RL, preference learning, human feedback, RLHF, reward learning, inverse RL, imitation learning, demonstrations, behavior cloning, robot manipulation, locomotion, D4RL, RoboMimic, BridgeData, Meta-World, Adroit, kitchen tasks, language-conditioned robot learning, and foundation models for robotics. Records were retained when title or abstract evidence connected robotics or robot learning with offline RL, preference or human-feedback RL, demonstrations, reward learning, safety, adaptation,

datasets, or evaluation. Duplicates were removed by arXiv identifier, and the final bibliography cites 300 unique scholarly preprints.

3.2 Taxonomy

We organize the field along six axes. The first is *data source*: expert demonstrations, mixed-quality logs, teleoperation, scripted controllers, simulation, self-supervised robot play, or web-scale robot datasets. The second is *optimization objective*: conservative value learning, behavior cloning, advantage-weighted regression, sequence modeling, model-based planning, or policy improvement from preferences. The third is *feedback type*: scalar rewards, pairwise preferences, rankings, corrections, language feedback, or implicit human choices. The fourth is *embodiment*: manipulation, dexterous control, locomotion, navigation, or mobile manipulation. The fifth is *deployment pathway*: offline-only, offline-to-online, sim-to-real, or human-in-the-loop refinement. The sixth is *risk control*: uncertainty, constraints, conservatism, safe exploration, and evaluation under distribution shift.

3.3 Conservative Offline RL Objectives

Conservative offline RL methods penalize out-of-distribution actions, constrain policy updates toward the behavior policy, or lower-bound value estimates so that learned policies do not exploit unsupported value errors. These techniques are central in robotics because small value overestimation can lead to unsafe contact-rich behavior. However, over-conservatism can prevent improvement over demonstrations, especially when datasets are narrow or contain suboptimal behavior [5, 14, 111, 131, 168, 204, 218, 219].

3.4 Sequence Models and Demonstration Learning

Robotic datasets often contain trajectories rather than independent transitions. Sequence models, behavior cloning, return-conditioned policies, and transformer-based controllers can exploit this structure. Demonstration learning remains important because expert or teleoperated trajectories provide high-quality support for manipulation and navigation. Offline RL extends imitation by improving beyond the data when rewards or value estimates are reliable [111, 131, 141, 165, 182, 196, 236, 299].

3.5 Preference and Reward Learning

Preference-based robot learning typically collects comparisons between trajectory segments, learns a reward model, and optimizes a policy under that learned reward. Variants use active query selection, uncertainty-aware preference sampling, language feedback, corrective demonstrations, or hybrid reward functions. The central difficulty is reward misspecification: a reward model can overfit annotator shortcuts or become unreliable outside the comparison distribution [5, 14, 111, 168, 204, 218, 236, 291].

3.6 Robot Manipulation, Locomotion, and Control

Manipulation tasks stress dexterity, contact dynamics, visual generalization, and object diversity. Locomotion tasks stress dynamics, robustness, and safety. Navigation tasks stress partial observability and long horizons. Offline and preference-based methods must therefore align algorithmic assumptions with task physics: a policy that is safe for tabletop manipulation may not be safe for locomotion, and a preference model trained on short clips may not capture long-horizon task success [111, 131, 165, 168, 204, 218, 236, 291].

3.7 Offline-to-Online and Sim-to-Real Adaptation

A practical robot-learning pipeline often begins offline and then improves online with limited real interaction. Offline-to-online methods use conservative initialization, uncertainty-aware exploration, or constrained policy updates so that the robot can improve without catastrophic failures. Sim-to-real methods use randomized simulation, real-world calibration, and adaptation to bridge dynamics and perception gaps [14, 49, 101, 141, 174, 227, 236, 267].

3.8 Benchmarks and Datasets

Benchmarks such as offline control suites, robotic manipulation datasets, imitation-learning corpora, and task-specific robot logs have shaped the field. Yet robotics evaluation remains difficult because simulators simplify contact, hardware varies, and real-world resets are costly. Useful benchmarks should report data quality, coverage, reward access, safety violations, offline-to-online protocols, and real-robot transfer rather than only final return [14, 111, 131, 138, 168, 204, 218, 236].

3.9 Foundation Models and Language-Conditioned Robots

Foundation models enter this field in several ways: they can condition robot policies on language, provide reward features, generate task proposals, summarize feedback, or serve as priors for generalist robot policies. Their use does not remove the offline RL problem; instead, it changes the data and reward interfaces. Language-conditioned robots still need grounded rewards, physical safety, and reliable adaptation under embodiment shift [14, 111, 138, 165, 168, 204, 218, 236].

3.10 Representative Literature Map

The following map cites the full retained corpus by theme, making the survey coverage explicit across offline RL, preference feedback, demonstrations, safety, benchmarks, and robot deployments.

Offline and batch reinforcement learning The offline and batch reinforcement learning cluster contains work that develops methods, systems, datasets, or analyses central to this survey [33, 34, 89, 154, 196, 242, 262, 281, 294]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [77, 97, 132, 166, 197, 211, 232, 249, 296]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [50, 59, 72, 76, 152, 195].

Preference-based and human-feedback reinforcement learning The preference-based and human-feedback reinforcement learning cluster contains work that develops methods, systems, datasets, or analyses central to this survey [29, 92, 141, 150, 182, 204, 220, 236, 291]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [30, 58, 96, 118, 157, 178, 185, 193, 213]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [22, 27, 57, 60, 73, 94, 190, 191, 285]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [127, 129, 135, 171, 234, 241, 248, 266, 284]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [23, 24, 44, 133, 183, 239, 250, 255, 282]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [54,

110, 187, 188, 198, 201, 203, 240, 259]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [18, 36, 116, 158, 179, 215, 243, 251, 297]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [1, 41, 95, 103, 200, 235, 245, 257, 274]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [47, 87, 105, 136, 170, 210, 247, 252].

Imitation learning and demonstrations The imitation learning and demonstrations cluster contains work that develops methods, systems, datasets, or analyses central to this survey [43, 51, 65, 202, 205, 258, 265, 289, 299]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [20, 93, 130, 140, 175, 180, 217, 270, 278]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [26, 276].

Robot manipulation and control The robot manipulation and control cluster contains work that develops methods, systems, datasets, or analyses central to this survey [101, 111, 131, 138, 165, 168, 218, 219, 221]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [53, 64, 124, 147, 155, 163, 227, 268, 283]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [3, 38, 39, 107, 122, 146, 174, 216, 231]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [28, 35, 37, 125, 156, 172, 207, 226, 261]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [11, 52, 68, 112, 117, 139, 177, 277, 290]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [15, 25, 86, 143, 167, 169, 238, 292, 298]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [6, 12, 21, 55, 88, 90, 189, 273, 275]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [46, 62, 69, 109, 144, 151, 212, 228, 246]. Additional retained studies broaden

the cluster with alternative architectures, training signals, data sources, and deployment assumptions [8, 56, 79, 91, 123, 148, 256, 272, 288]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [17, 32, 45, 134, 181, 229, 230, 264, 300]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [9, 31, 61, 71, 102, 115, 126, 128, 164]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [70, 74, 80, 153, 161, 224, 244, 287, 295]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [2, 7, 121, 206, 209, 214, 222, 263, 286]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [10, 63, 66, 67, 75, 100, 137, 142, 233]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [42, 85, 162, 184, 186, 254, 260, 271, 279]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [13, 16, 19, 83, 84, 114, 119, 160, 199].

Offline-to-online and sim-to-real adaptation The offline-to-online and sim-to-real adaptation cluster contains work that develops methods, systems, datasets, or analyses central to this survey [49, 225, 293].

Safety, uncertainty, and conservative objectives The safety, uncertainty, and conservative objectives cluster contains work that develops methods, systems, datasets, or analyses central to this survey [40, 48, 78, 81, 99, 106, 120, 145, 192]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [113].

Datasets, benchmarks, and evaluation The datasets, benchmarks, and evaluation cluster contains work that develops methods, systems, datasets, or analyses central to this survey [4, 14, 98, 104, 173, 176, 237, 267, 280]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [82, 108, 149, 159, 194, 208, 223, 269].

Foundation models and language-conditioned robot learning The foundation models and language-conditioned robot learning cluster contains work that develops methods, systems, datasets, or analyses central to this survey [5, 253].

4 Challenges and Prospects

4.1 Dataset Coverage and Hidden Bias

Offline robot policies inherit the coverage and bias of their datasets. A dataset may omit rare failures, hard objects, unusual contacts, or recovery behaviors. Future methods need coverage diagnostics that predict when a policy is leaving the support of its data and when online intervention or human feedback is required.

4.2 Reward Model Reliability

Preference-based methods depend on reward models that generalize beyond labeled comparisons. In robotics, this is risky because a small reward error can produce unsafe physical behavior. Reward learning should therefore be paired with uncertainty estimates, adversarial evaluation, active queries, and conservative policy optimization.

4.3 Safe Offline-to-Online Improvement

The most useful robot systems will not remain purely offline. They will start from offline data, ask humans for feedback, and improve through limited online interaction. The challenge is to bound risk during improvement while still allowing policies to surpass demonstrations.

4.4 Human Feedback Interfaces

Human feedback is not free. Query design, annotator burden, explanation, disagreement, and timing all affect preference quality. Future robotics systems should make feedback collection ergonomic: the robot should ask useful questions, expose uncertainty, and translate preferences into verifiable policy changes.

4.5 Toward Integrated Robot-Learning Pipelines

Offline data, demonstrations, preference labels, learned rewards, foundation-model priors, and real-world fine-tuning should be treated as parts of a single pipeline. Progress will depend on reproducible

datasets, transparent reward models, safety-aware optimization, and evaluation protocols that include physical deployment constraints.

5 Conclusion

Offline and preference-based reinforcement learning offer a practical route for robot learning under limited interaction and imperfect reward specification. Offline RL learns from logged experience; preference-based RL learns what humans value; imitation learning and demonstrations provide structured behavior; and offline-to-online adaptation connects static datasets to deployment. This survey reviewed the methods, datasets, tasks, safety concerns, and prospects that define the area. The central lesson is that robot learning should jointly reason about data support, human judgment, uncertainty, and physical risk. Systems that integrate these pieces will be better positioned to learn useful robot behavior without relying on unsafe trial-and-error.

References

- [1] Bogdan Felician Abaza, Andrei-Alexandru Staicu, and Cristian Vasile Doicin. A Semantic Autonomy Framework for VLM-Integrated Indoor Mobile Robots: Hybrid Deterministic Reasoning and Cross-Robot Adaptive Memory. *arXiv preprint arXiv:2605.02525*, 2026. URL <https://arxiv.org/abs/2605.02525>.
- [2] Haruki Abe, Takayuki Osa, Yusuke Mukuta, and Tatsuya Harada. Cross-Embodiment Offline Reinforcement Learning for Heterogeneous Robot Datasets. *arXiv preprint arXiv:2602.18025*, 2026. URL <https://arxiv.org/abs/2602.18025>.
- [3] Suzan Ece Ada, Erhan Oztog, and Emre Ugur. Diffusion Policies for Out-of-Distribution Generalization in Offline Reinforcement Learning. *arXiv preprint arXiv:2307.04726*, 2023. URL <https://arxiv.org/abs/2307.04726>.
- [4] Ameya Agaskar, Sriram Siva, William Pickering, Kyle O’Brien, Charles Kekeh, Alexandre Ormiga Galvao Barbosa, Ang Li, Brianna Gallo Sarker, Alicia Chua, Mayur Nemade, Charun Thattai, Jiaming Di, Isaac Iyengar, Ramya Dharoor, Dino Kirouani, Jimmy Erskine, Tamir Hegazy, Scott Niekum, Usman A. Khan, Federico Pecora, et al. DeepFleet: Multi-Agent Foundation Models for Mobile Robots. *arXiv preprint arXiv:2508.08574*, 2025. URL <https://arxiv.org/abs/2508.08574>.
- [5] Michael Ahn, Debidatta Dwibedi, Chelsea Finn, Montse Gonzalez Arenas, Keerthana Gopalakrish-

- nan, Karol Hausman, Brian Ichter, Alex Irpan, Nikhil Joshi, Ryan Julian, Sean Kirmani, Isabel Leal, Edward Lee, Sergey Levine, Yao Lu, Isabel Leal, Sharath Maddineni, Kanishka Rao, Dorsa Sadigh, Pannag Sanketi, et al. AutoRT: Embodied Foundation Models for Large Scale Orchestration of Robotic Agents. *arXiv preprint arXiv:2401.12963*, 2024. URL <https://arxiv.org/abs/2401.12963>.
- [6] Orhan Eren Akgun, Nestor Cuevas, Matheus Farias, and Daniel Garces. Tiny Reinforcement Learning for Quadruped Locomotion using Decision Transformers. *arXiv preprint arXiv:2402.13201*, 2024. URL <https://arxiv.org/abs/2402.13201>.
- [7] Riad Akrou, Marc Schoenauer, and Michele Sebag. APRIL: Active Preference-learning based Reinforcement Learning. *arXiv preprint arXiv:1208.0984*, 2012. URL <https://arxiv.org/abs/1208.0984>.
- [8] Minttu Alakuijala, Reginald McLean, Isaac Woungang, Nariman Farsad, Samuel Kaski, Pekka Marttinen, and Kai Yuan. Video-Language Critic: Transferable Reward Functions for Language-Conditioned Robotics. *arXiv preprint arXiv:2405.19988*, 2024. URL <https://arxiv.org/abs/2405.19988>.
- [9] Mohammad Saeid Anwar, Anuradha Ravi, Indrajeet Ghosh, Gaurav Shinde, Carl Busart, and Nirmalya Roy. COHORT: Hybrid RL for Collaborative Large DNN Inference on Multi-Robot Systems Under Real-Time Constraints. *arXiv preprint arXiv:2603.10436*, 2026. URL <https://arxiv.org/abs/2603.10436>.
- [10] Riku Arakawa, Sosuke Kobayashi, Yuya Unno, Yuta Tsuboi, and Shin ichi Maeda. DQN-TAMER: Human-in-the-Loop Reinforcement Learning with Intractable Feedback. *arXiv preprint arXiv:1810.11748*, 2018. URL <https://arxiv.org/abs/1810.11748>.
- [11] Merve Atasever, Matthew Hong, Mihir Nitin Kulkarni, Qingpei Li, and Jyotirmoy V. Deshmukh. Multi-Agent Path Finding via Offline RL and LLM Collaboration. *arXiv preprint arXiv:2509.22130*, 2025. URL <https://arxiv.org/abs/2509.22130>.
- [12] Anl Can Ates, Orhan Kahraman, and Cihan Topal. RMTL: Reinforced Micro-task Learning for Long-Horizon Manipulation with VLM Rewards. *arXiv preprint arXiv:2606.26175*, 2026. URL <https://arxiv.org/abs/2606.26175>.
- [13] Shingo Ayabe, Hiroshi Kera, and Kazuhiko Kawamoto. Adversarial Fine-tuning in Offline-to-Online Reinforcement Learning for Robust Robot Control. *arXiv preprint arXiv:2510.13358*, 2025. URL <https://arxiv.org/abs/2510.13358>.
- [14] Ryan Bahlous-Boldi, Li Ding, Lee Spector, and Scott Niekum. Pareto-Optimal Learning from Preferences with Hidden Context. *arXiv preprint arXiv:2406.15599*, 2024. URL <https://arxiv.org/abs/2406.15599>.
- [15] Fengshuo Bai, Runze Liu, Yali Du, Ying Wen, and Yaodong Yang. RAT: Adversarial Attacks on Deep Reinforcement Agents for Targeted Behaviors. *arXiv preprint arXiv:2412.10713*, 2024. URL <https://arxiv.org/abs/2412.10713>.
- [16] Daulet Baimukashev, Gokhan Alcan, Kevin Sebastian Luck, and Ville Kyrki. Learning Transparent Reward Models via Unsupervised Feature Selection. *arXiv preprint arXiv:2410.18608*, 2024. URL <https://arxiv.org/abs/2410.18608>.
- [17] Max Balsells, Marcel Torne, Zihan Wang, Samedh Desai, Pulkit Agrawal, and Abhishek Gupta. Autonomous Robotic Reinforcement Learning with Asynchronous Human Feedback. *arXiv preprint arXiv:2310.20608*, 2023. URL <https://arxiv.org/abs/2310.20608>.
- [18] Shreyas Bhat, Joseph B. Lyons, Cong Shi, and X. Jessie Yang. Effect of Adapting to Human Preferences on Trust in Human-Robot Teaming. *arXiv preprint arXiv:2309.05179*, 2023. URL <https://arxiv.org/abs/2309.05179>.
- [19] Chethan Bhateja, Derek Guo, Dibya Ghosh, Anikait Singh, Manan Tomar, Quan Vuong, Yevgen Chebotar, Sergey Levine, and Aviral Kumar. Robotic Offline RL from Internet Videos via Value-Function Pre-Training. *arXiv preprint arXiv:2309.13041*, 2023. URL <https://arxiv.org/abs/2309.13041>.
- [20] Ondrej Biza, Thomas Weng, Lingfeng Sun, Karl Schmeckpeper, Tarik Kelestemur, Yecheng Jason Ma, Robert Platt, Jan-Willem van de Meent, and Lawson L. S. Wong. On-Robot Reinforcement Learning with Goal-Contrastive Rewards. *arXiv preprint arXiv:2410.19989*, 2024. URL <https://arxiv.org/abs/2410.19989>.
- [21] Konstantinos Bousmalis, Giulia Vezzani, Dushyant Rao, Coline Devin, Alex X. Lee, Maria Bauza, Todor Davchev, Yuxiang Zhou, Agrim Gupta, Akhil Raju, Antoine Laurens, Claudio Fantacci, Valentin Dalibard, Martina Zambelli, Murilo Martins, Rugile Pevceviciute, Michiel Blokzijl, Misha Denil, Nathan Batchelor, Thomas Lampe, et al. RoboCat: A Self-Improving Generalist Agent for Robotic Manipulation. *arXiv preprint arXiv:2306.11706*, 2023. URL <https://arxiv.org/abs/2306.11706>.
- [22] Alexander Bukharin, Ilgee Hong, Haoming Jiang, Zichong Li, Qingru Zhang, Zixuan Zhang, and Tuo Zhao. Robust Reinforcement Learning from Corrupted Human Feedback. *arXiv preprint*

- arXiv:2406.15568*, 2024. URL <https://arxiv.org/abs/2406.15568>.
- [23] Erdem Byk. Learning Preferences for Interactive Autonomy. *arXiv preprint arXiv:2210.10899*, 2022. URL <https://arxiv.org/abs/2210.10899>.
- [24] Erdem Byk, Nicolas Huynh, Mykel J. Kochenderfer, and Dorsa Sadigh. Active Preference-Based Gaussian Process Regression for Reward Learning. *arXiv preprint arXiv:2005.02575*, 2020. URL <https://arxiv.org/abs/2005.02575>.
- [25] Serkan Cabi, Sergio Gomez Colmenarejo, Alexander Novikov, Ksenia Konyushkova, Scott Reed, Rae Jeong, Konrad Zolna, Yusuf Aytar, David Budden, Mel Vecerik, Oleg Sushkov, David Barker, Jonathan Scholz, Misha Denil, Nando de Freitas, and Ziyu Wang. Scaling data-driven robotics with reward sketching and batch reinforcement learning. *arXiv preprint arXiv:1909.12200*, 2019. URL <https://arxiv.org/abs/1909.12200>.
- [26] Haoyuan Cai, Zhenghao Peng, and Bolei Zhou. Predictive Preference Learning from Human Interventions. *arXiv preprint arXiv:2510.01545*, 2025. URL <https://arxiv.org/abs/2510.01545>.
- [27] Chenyang Cao, Miguel Rogel-Garcia, Mohamed Nabail, Xueqian Wang, and Nicholas Rhinehart. Residual Reward Models for Preference-based Reinforcement Learning. *arXiv preprint arXiv:2507.00611*, 2025. URL <https://arxiv.org/abs/2507.00611>.
- [28] Federico Ceola, Lorenzo Natale, Niko Sunderhauf, and Krishan Rana. LHManip: A Dataset for Long-Horizon Language-Grounded Manipulation Tasks in Cluttered Tabletop Environments. *arXiv preprint arXiv:2312.12036*, 2023. URL <https://arxiv.org/abs/2312.12036>.
- [29] Souradip Chakraborty, Anukriti Singh, Amisha Bhaskar, Pratap Tokekar, Dinesh Manocha, and Amrit Singh Bedi. REBEL: Reward Regularization-Based Approach for Robotic Reinforcement Learning from Human Feedback. *arXiv preprint arXiv:2312.14436*, 2023. URL <https://arxiv.org/abs/2312.14436>.
- [30] Souradip Chakraborty, Jiahao Qiu, Hui Yuan, Alec Koppel, Furong Huang, Dinesh Manocha, Amrit Singh Bedi, and Mengdi Wang. MaxMin-RLHF: Alignment with Diverse Human Preferences. *arXiv preprint arXiv:2402.08925*, 2024. URL <https://arxiv.org/abs/2402.08925>.
- [31] Bryan Chan, Anson Leung, and James Bergstra. Offline-to-online Reinforcement Learning for Image-based Grasping with Scarce Demonstrations. *arXiv preprint arXiv:2410.14957*, 2024. URL <https://arxiv.org/abs/2410.14957>.
- [32] Wei-Di Chang, Juan Camilo Gamboa Higuera, Scott Fujimoto, David Meger, and Gregory Dudek. IL-floW: Imitation Learning from Observation using Normalizing Flows. *arXiv preprint arXiv:2205.09251*, 2022. URL <https://arxiv.org/abs/2205.09251>.
- [33] Annie S. Chen, HyunJi Nam, Suraj Nair, and Chelsea Finn. Batch Exploration with Examples for Scalable Robotic Reinforcement Learning. *arXiv preprint arXiv:2010.11917*, 2020. URL <https://arxiv.org/abs/2010.11917>.
- [34] Chang Chen, Yuecheng Liu, Yuzheng Zhuang, Sitong Mao, Kaiwen Xue, and Shunbo Zhou. SCALE: Self-Correcting Visual Navigation for Mobile Robots via Anti-Noveltly Estimation. *arXiv preprint arXiv:2404.10675*, 2024. URL <https://arxiv.org/abs/2404.10675>.
- [35] Hongyu Chen and Guangrun Wang. UML-CoT: Structured Reasoning and Planning with Unified Modeling Language for Robotic Room Cleaning. *arXiv preprint arXiv:2509.22628*, 2025. URL <https://arxiv.org/abs/2509.22628>.
- [36] Kuan-Chen Chen, Winston Chen, Wei-Fang Sun, and Min-Chun Hu. VLM-AR3L: Vision-Language Models for Absolute and Relative Rewards in Reinforcement Learning. *arXiv preprint arXiv:2607.00483*, 2026. URL <https://arxiv.org/abs/2607.00483>.
- [37] Qianzhong Chen, Justin Yu, Mac Schwager, Pieter Abbeel, Yide Shentu, and Philipp Wu. SARM: Stage-Aware Reward Modeling for Long Horizon Robot Manipulation. *arXiv preprint arXiv:2509.25358*, 2025. URL <https://arxiv.org/abs/2509.25358>.
- [38] Qianzhong Chen, Hau Zheng, Justin Yu, Suning Huang, Jiankai Sun, Ken Goldberg, Chuan Wen, Pieter Abbeel, Yide Shentu, Philipp Wu, and Mac Schwager. SARM2: Multi-Task Stage Aware Reward Modeling for Self Improving Robotic Manipulation. *arXiv preprint arXiv:2606.10305*, 2026. URL <https://arxiv.org/abs/2606.10305>.
- [39] Shirui Chen, Cole Harrison, Ying-Chun Lee, Angela Jin Yang, Zhongzheng Ren, Lillian J. Ratliff, Jiafei Duan, Dieter Fox, and Ranjay Krishna. TOPReward: Token Probabilities as Hidden Zero-Shot Rewards for Robotics. *arXiv preprint arXiv:2602.19313*, 2026. URL <https://arxiv.org/abs/2602.19313>.
- [40] Timothy Chen, Marcus Dominguez-Kuhne, Aiden Swann, Xu Liu, and Mac Schwager. Semantic-Metric Bayesian Risk Fields: Learning Robot Safety from Human Videos with a VLM Prior. *arXiv preprint arXiv:2512.08233*, 2025. URL <https://arxiv.org/abs/2512.08233>.
- [41] Wendi Chen, Han Xue, Fangyuan Zhou, Yuan Fang, and Cewu Lu. DeformPAM: Data-Efficient

- Learning for Long-horizon Deformable Object Manipulation via Preference-based Action Alignment. *arXiv preprint arXiv:2410.11584*, 2024. URL <https://arxiv.org/abs/2410.11584>.
- [42] Xi Chen, Ali Ghadirzadeh, Tianhe Yu, Yuan Gao, Jianhao Wang, Wenzhe Li, Bin Liang, Chelsea Finn, and Chongjie Zhang. Latent-Variable Advantage-Weighted Policy Optimization for Offline RL. *arXiv preprint arXiv:2203.08949*, 2022. URL <https://arxiv.org/abs/2203.08949>.
- [43] Xiangyu Chen, Chuhao Zhou, Yuxi Liu, and Jianfei Yang. RM-RL: Role-Model Reinforcement Learning for Precise Robot Manipulation. *arXiv preprint arXiv:2510.15189*, 2025. URL <https://arxiv.org/abs/2510.15189>.
- [44] Xiyue Chen, Muhan Lin, Shuyang Shi, and Joseph Campbell. MAPL: Multi-Objective Preference Learning for Robot Locomotion. *arXiv preprint arXiv:2606.25398*, 2026. URL <https://arxiv.org/abs/2606.25398>.
- [45] Yiwen Chen, Yuyao Ye, Ziyi Chen, Chuheng Zhang, and Marcelo H. Ang. ARO: Large Language Model Supervised Robotics Text2Skill Autonomous Learning. *arXiv preprint arXiv:2403.15834*, 2024. URL <https://arxiv.org/abs/2403.15834>.
- [46] Yuxin Chen, Devesh K. Jha, Masayoshi Tomizuka, and Diego Romeres. FDPP: Fine-tune Diffusion Policy with Human Preference. *arXiv preprint arXiv:2501.08259*, 2025. URL <https://arxiv.org/abs/2501.08259>.
- [47] Jie Cheng, Gang Xiong, Xingyuan Dai, Qinghai Miao, Yisheng Lv, and Fei-Yue Wang. RIME: Robust Preference-based Reinforcement Learning with Noisy Preferences. *arXiv preprint arXiv:2402.17257*, 2024. URL <https://arxiv.org/abs/2402.17257>.
- [48] Minjae Cho and Chuangchuang Sun. Sparsity-based Safety Conservatism for Constrained Offline Reinforcement Learning. *arXiv preprint arXiv:2407.13006*, 2024. URL <https://arxiv.org/abs/2407.13006>.
- [49] Andrew Choi and Wei Xu. RankQ: Offline-to-Online Reinforcement Learning via Self-Supervised Action Ranking. *arXiv preprint arXiv:2605.11151*, 2026. URL <https://arxiv.org/abs/2605.11151>.
- [50] Soon Jynn Chu, Raju Gottumukkala, and Alan Barhorst. Towards an Adaptive Social Game-Playing Robot: An Offline Reinforcement Learning-Based Framework. *arXiv preprint arXiv:2509.16858*, 2025. URL <https://arxiv.org/abs/2509.16858>.
- [51] Caleb Chuck, Carl Qi, Michael J. Munje, Shuoze Li, Max Rudolph, Chang Shi, Siddhant Agarwal, Harshit Sikchi, Abhinav Peri, Sarthak Dayal, Evan Kuo, Kavan Mehta, Anthony Wang, Peter Stone, Amy Zhang, and Scott Niekum. Robot Air Hockey: A Manipulation Testbed for Robot Learning with Reinforcement Learning. *arXiv preprint arXiv:2405.03113*, 2024. URL <https://arxiv.org/abs/2405.03113>.
- [52] Jaden Clark, Joey Hejna, and Dorsa Sadigh. Efficiently Generating Expressive Quadruped Behaviors via Language-Guided Preference Learning. *arXiv preprint arXiv:2502.03717*, 2025. URL <https://arxiv.org/abs/2502.03717>.
- [53] Nicholas E. Corrado, Yuxiao Qu, John U. Balis, Adam Labiosa, and Josiah P. Hanna. Guided Data Augmentation for Offline Reinforcement Learning and Imitation Learning. *arXiv preprint arXiv:2310.18247*, 2023. URL <https://arxiv.org/abs/2310.18247>.
- [54] Peter Crowley, Zachary Serlin, Tyler Paine, Makai Mann, Michael Benjamin, and Calin Belta. SPLASH! Sample-efficient Preference-based inverse reinforcement learning for Long-horizon Adversarial tasks from Suboptimal Hierarchical demonstrations. *arXiv preprint arXiv:2507.08707*, 2025. URL <https://arxiv.org/abs/2507.08707>.
- [55] Yuchen Cui, Qiping Zhang, Alessandro Allievi, Peter Stone, Scott Niekum, and W. Bradley Knox. The EMPATHIC Framework for Task Learning from Implicit Human Feedback. *arXiv preprint arXiv:2009.13649*, 2020. URL <https://arxiv.org/abs/2009.13649>.
- [56] Mingtong Dai, Lingbo Liu, Yongjie Bai, Yang Liu, Zhouxia Wang, Rui SU, Chunjie Chen, Liang Lin, and Xinyu Wu. RoVer: Robot Reward Model as Test-Time Verifier for Vision-Language-Action Model. *arXiv preprint arXiv:2510.10975*, 2025. URL <https://arxiv.org/abs/2510.10975>.
- [57] Jorge de Heuvel, Florian Seiler, and Maren Bennewitz. EnQuery: Ensemble Policies for Diverse Query-Generation in Preference Alignment of Robot Navigation. *arXiv preprint arXiv:2404.04852*, 2024. URL <https://arxiv.org/abs/2404.04852>.
- [58] Jorge de Heuvel, Daniel Marta, Simon Holk, Iolanda Leite, and Maren Bennewitz. The Impact of VR and 2D Interfaces on Human Feedback in Preference-Based Robot Learning. *arXiv preprint arXiv:2503.16500*, 2025. URL <https://arxiv.org/abs/2503.16500>.
- [59] Christopher Diehl, Timo Sievernich, Martin Kruger, Frank Hoffmann, and Torsten Bertram. UMBRELLA: Uncertainty-Aware Model-Based Offline Reinforcement Learning Leveraging Planning. *arXiv preprint arXiv:2111.11097*, 2021. URL <https://arxiv.org/abs/2111.11097>.

- [60] Li Ding, Jenny Zhang, Jeff Clune, Lee Spector, and Joel Lehman. Quality Diversity through Human Feedback: Towards Open-Ended Diversity-Driven Optimization. *arXiv preprint arXiv:2310.12103*, 2023. URL <https://arxiv.org/abs/2310.12103>.
- [61] Wenhao Ding, Nathalie Majcherczyk, Mohit Deshpande, Xuewei Qi, Ding Zhao, Rajasimman Mahivanan, and Arnie Sen. Learning to View: Decision Transformers for Active Object Detection. *arXiv preprint arXiv:2301.09544*, 2023. URL <https://arxiv.org/abs/2301.09544>.
- [62] Zihan Ding, Yuanpei Chen, Allen Z. Ren, Shixiang Shane Gu, Qianxu Wang, Hao Dong, and Chi Jin. Learning a Universal Human Prior for Dexterous Manipulation from Human Preference. *arXiv preprint arXiv:2304.04602*, 2023. URL <https://arxiv.org/abs/2304.04602>.
- [63] Perry Dong, Suvir Mirchandani, Dorsa Sadigh, and Chelsea Finn. What Matters for Batch Online Reinforcement Learning in Robotics? *arXiv preprint arXiv:2505.08078*, 2025. URL <https://arxiv.org/abs/2505.08078>.
- [64] Yujian Dong, Tianyu Wu, and Chaoyang Song. Optimizing Robotic Manipulation with Decision-RWKV: A Recurrent Sequence Modeling Approach for Lifelong Learning. *arXiv preprint arXiv:2407.16306*, 2024. URL <https://arxiv.org/abs/2407.16306>.
- [65] Christian Ellis, Maggie Wigness, John G. Rogers, Craig Lennon, and Lance Fiondella. Risk Averse Bayesian Reward Learning for Autonomous Navigation from Human Demonstration. *arXiv preprint arXiv:2108.00276*, 2021. URL <https://arxiv.org/abs/2108.00276>.
- [66] Patrick Emedom-Nnamdi, Abram L. Friesen, Bobak Shahriari, Nando de Freitas, and Matt W. Hoffman. Knowledge Transfer from Teachers to Learners in Growing-Batch Reinforcement Learning. *arXiv preprint arXiv:2305.03870*, 2023. URL <https://arxiv.org/abs/2305.03870>.
- [67] Nicolas Espinosa-Dice, Kianté Brantley, and Wen Sun. Expressive Value Learning for Scalable Offline Reinforcement Learning. *arXiv preprint arXiv:2510.08218*, 2025. URL <https://arxiv.org/abs/2510.08218>.
- [68] Edoardo Fazzari, Omar Mohamed, Khalfan Hableel, Hamdan Alhadhrami, and Cesare Stefanini. Hybrid Offline-Online Reinforcement Learning for Sensorless, High-Precision Force Regulation in Surgical Robotic Grasping. *arXiv preprint arXiv:2602.23870*, 2026. URL <https://arxiv.org/abs/2602.23870>.
- [69] Youhe Feng, Hansen Shi, Haoyang Li, Xinlei Guo, Yang Wang, Chengyang Zhang, Jinkai Zhang, Xiaohan Zhang, Jie Tang, and Jing Zhang. ProcVLM: Learning Procedure-Grounded Progress Rewards for Robotic Manipulation. *arXiv preprint arXiv:2605.08774*, 2026. URL <https://arxiv.org/abs/2605.08774>.
- [70] Jiawei Fu, Yonghao Long, Kai Chen, Wang Wei, and Qi Dou. Multi-objective Cross-task Learning via Goal-conditioned GPT-based Decision Transformers for Surgical Robot Task Automation. *arXiv preprint arXiv:2405.18757*, 2024. URL <https://arxiv.org/abs/2405.18757>.
- [71] Keisuke Fujii, Kazushi Tsutsui, Yu Teshima, Makoto Itoh, Naoya Takeishi, Nozomi Nishiumi, Ryoya Tanaka, Shunsuke Shigaki, and Yoshinobu Kawahara. Data-driven simulator of multi-animal behavior with unknown dynamics via offline and online reinforcement learning. *arXiv preprint arXiv:2510.10451*, 2025. URL <https://arxiv.org/abs/2510.10451>.
- [72] Pawe Gajewski, Dominik Zurek, Marcin Pietron, and Kamil Faber. Goal-Conditioned Decision Transformer for Multi-Goal Offline Reinforcement Learning. *arXiv preprint arXiv:2410.06347*, 2024. URL <https://arxiv.org/abs/2410.06347>.
- [73] Xiancheng Gao, Yufeng Shi, Wengang Zhou, and Houqiang Li. Search-Based Credit Assignment for Offline Preference-Based Reinforcement Learning. *arXiv preprint arXiv:2508.15327*, 2025. URL <https://arxiv.org/abs/2508.15327>.
- [74] Divyansh Garg, Skanda Vaidyanath, Kuno Kim, Jiaming Song, and Stefano Ermon. LISA: Learning Interpretable Skill Abstractions from Language. *arXiv preprint arXiv:2203.00054*, 2022. URL <https://arxiv.org/abs/2203.00054>.
- [75] Koffivi Fidele Gbagbe, Miguel Altamirano Cabrera, Ali Alabbas, Oussama Alyunes, Artem Lykov, and Dzmitry Tsetserukou. Bi-VLA: Vision-Language-Action Model-Based System for Bimanual Robotic Dexterous Manipulations. *arXiv preprint arXiv:2405.06039*, 2024. URL <https://arxiv.org/abs/2405.06039>.
- [76] Abdelghani Ghanem, Philippe Ciblat, and Mounir Ghogho. Multi-Objective Decision Transformers for Offline Reinforcement Learning. *arXiv preprint arXiv:2308.16379*, 2023. URL <https://arxiv.org/abs/2308.16379>.
- [77] Chen Gong, Zhou Yang, Yunpeng Bai, Junda He, Jieke Shi, Kecen Li, Arunesh Sinha, Bowen Xu, Xinwen Hou, David Lo, and Tianhao Wang. BAFFLE: Hiding Backdoors in Offline Reinforcement Learning Datasets. *arXiv preprint arXiv:2210.04688*, 2022. URL <https://arxiv.org/abs/2210.04688>.
- [78] Antonio Guillen-Perez. Mining the Long Tail: A Comparative Study of Data-Centric Criticality Metrics for Robust Offline Reinforcement Learning

- in Autonomous Motion Planning. *arXiv preprint arXiv:2508.18397*, 2025. URL <https://arxiv.org/abs/2508.18397>.
- [79] Christian Gumbsch, Leonardo Barcellona, Lennard Schunemann, Platon Karageorgis, Andrii Zadaianchuk, Zehao Wang, Sergey Zakharov, Fabien Despinoy, Rahaf Aljundi, and Efstratios Gavves. From Demonstrations to Rewards: Test-Time Prompt Optimization for VLM Reward Models. *arXiv preprint arXiv:2606.00083*, 2026. URL <https://arxiv.org/abs/2606.00083>.
- [80] Dingkun Guo, Yuqi Xiang, Shuqi Zhao, Xinghao Zhu, Masayoshi Tomizuka, Mingyu Ding, and Wei Zhan. PhyGrasp: Generalizing Robotic Grasping with Physics-informed Large Multimodal Models. *arXiv preprint arXiv:2402.16836*, 2024. URL <https://arxiv.org/abs/2402.16836>.
- [81] Junyu Guo, Zhi Zheng, Donghao Ying, Ming Jin, Shangding Gu, Costas Spanos, and Javad Lavaei. Don’t Trade Off Safety: Diffusion Regularization for Constrained Offline RL. *arXiv preprint arXiv:2502.12391*, 2025. URL <https://arxiv.org/abs/2502.12391>.
- [82] Yihong Guo, Yu Yang, Pan Xu, and Anqi Liu. MOBODY: Model Based Off-Dynamics Offline Reinforcement Learning. *arXiv preprint arXiv:2506.08460*, 2025. URL <https://arxiv.org/abs/2506.08460>.
- [83] Gunshi Gupta, Tim G. J. Rudner, Rowan Thomas McAllister, Adrien Gaidon, and Yarin Gal. Can Active Sampling Reduce Causal Confusion in Offline Reinforcement Learning? *arXiv preprint arXiv:2312.17168*, 2023. URL <https://arxiv.org/abs/2312.17168>.
- [84] Nico Gurtler, Sebastian Blaes, Pavel Kolev, Felix Widmaier, Manuel Wuthrich, Stefan Bauer, Bernhard Scholkopf, and Georg Martius. Benchmarking Offline Reinforcement Learning on Real-Robot Hardware. *arXiv preprint arXiv:2307.15690*, 2023. URL <https://arxiv.org/abs/2307.15690>.
- [85] Nico Gurtler, Felix Widmaier, Cansu Sancaktar, Sebastian Blaes, Pavel Kolev, Stefan Bauer, Manuel Wuthrich, Markus Wulfmeier, Martin Riedmiller, Arthur Allshire, Qiang Wang, Robert McCarthy, Hangeol Kim, Jongchan Baek, Wookyong Kwon, Shanliang Qian, Yasunori Toshimitsu, Mike Yan Michelis, Amirhossein Kazemipour, Arman Raayatsanati, et al. Real Robot Challenge 2022: Learning Dexterous Manipulation from Offline Data in the Real World. *arXiv preprint arXiv:2308.07741*, 2023. URL <https://arxiv.org/abs/2308.07741>.
- [86] Dongge Han, Trevor McInroe, Adam Jelley, Stefano V. Albrecht, Peter Bell, and Amos Storkey. LLM-Personalize: Aligning LLM Planners with Human Preferences via Reinforced Self-Training for Housekeeping Robots. *arXiv preprint arXiv:2404.14285*, 2024. URL <https://arxiv.org/abs/2404.14285>.
- [87] Cory Hayes and Matthew Marge. Towards Preference Learning for Autonomous Ground Robot Navigation Tasks. *arXiv preprint arXiv:2010.16361*, 2020. URL <https://arxiv.org/abs/2010.16361>.
- [88] Honglin He, Zhizheng Liu, Yukai Ma, and Bolei Zhou. From Imitation to Alignment: Human-Preference Flow Policies for Long-Horizon Sidewalk Navigation. *arXiv preprint arXiv:2606.12603*, 2026. URL <https://arxiv.org/abs/2606.12603>.
- [89] Linjin He, Xinda Qi, Dong Chen, Zhaojian Li, and Xiaobo Tan. DiSA-IQL: Offline Reinforcement Learning for Robust Soft Robot Control under Distribution Shifts. *arXiv preprint arXiv:2510.00358*, 2025. URL <https://arxiv.org/abs/2510.00358>.
- [90] Joey Hejna and Dorsa Sadigh. Few-Shot Preference Learning for Human-in-the-Loop RL. *arXiv preprint arXiv:2212.03363*, 2022. URL <https://arxiv.org/abs/2212.03363>.
- [91] Joey Hejna and Dorsa Sadigh. Inverse Preference Learning: Preference-based RL without a Reward Function. *arXiv preprint arXiv:2305.15363*, 2023. URL <https://arxiv.org/abs/2305.15363>.
- [92] Joey Hejna, Rafael Rafailov, Harshit Sikchi, Chelsea Finn, Scott Niekum, W. Bradley Knox, and Dorsa Sadigh. Contrastive Preference Learning: Learning from Human Feedback without RL. *arXiv preprint arXiv:2310.13639*, 2023. URL <https://arxiv.org/abs/2310.13639>.
- [93] Minh Heo, Youngwoon Lee, Doohyun Lee, and Joseph J. Lim. FurnitureBench: Reproducible Real-World Benchmark for Long-Horizon Complex Manipulation. *arXiv preprint arXiv:2305.12821*, 2023. URL <https://arxiv.org/abs/2305.12821>.
- [94] Ayano Hiranaka, Minjune Hwang, Sharon Lee, Chen Wang, Li Fei-Fei, Jiajun Wu, and Ruohan Zhang. Primitive Skill-based Robot Learning from Human Evaluative Feedback. *arXiv preprint arXiv:2307.15801*, 2023. URL <https://arxiv.org/abs/2307.15801>.
- [95] Simon Holk, Daniel Marta, and Iolanda Leite. PREDILECT: Preferences Delineated with Zero-Shot Language-based Reasoning in Reinforcement Learning. *arXiv preprint arXiv:2402.15420*, 2024. URL <https://arxiv.org/abs/2402.15420>.
- [96] Ilgee Hong, Zichong Li, Alexander Bukharin, Yixiao Li, Haoming Jiang, Tianbao Yang, and Tuo Zhao. Adaptive Preference Scaling for Reinforcement Learning with Human Feedback. *arXiv preprint arXiv:2406.02764*, 2024. URL <https://arxiv.org/abs/2406.02764>.

- [97] Jumman Hossain, Abu-Zaher Faridee, Nirmalya Roy, Jade Freeman, Timothy Gregory, and Theron T. Trout. EnCoMP: Enhanced Covert Maneuver Planning with Adaptive Threat-Aware Visibility Estimation using Offline Reinforcement Learning. *arXiv preprint arXiv:2403.20016*, 2024. URL <https://arxiv.org/abs/2403.20016>.
- [98] Yiwen Hou, Haoyuan Sun, Jinming Ma, and Feng Wu. Improving Offline Reinforcement Learning with Inaccurate Simulators. *arXiv preprint arXiv:2405.04307*, 2024. URL <https://arxiv.org/abs/2405.04307>.
- [99] Chao Huang, Wenshuo Zang, Carlo Pinciroli, Zhi Jane Li, Taposh Banerjee, Lili Su, and Rui Liu. Reactive Multi-Robot Navigation in Outdoor Environments Through Uncertainty-Aware Active Learning of Human Preference Landscape. *arXiv preprint arXiv:2409.16577*, 2024. URL <https://arxiv.org/abs/2409.16577>.
- [100] Dongchi Huang, Zhirui Fang, Tianle Zhang, Yihang Li, Lin Zhao, and Chunhe Xia. CO-RFT: Efficient Fine-Tuning of Vision-Language-Action Models through Chunked Offline Reinforcement Learning. *arXiv preprint arXiv:2508.02219*, 2025. URL <https://arxiv.org/abs/2508.02219>.
- [101] Haidong Huang, Haiyue Zhu, Jiayu Song, Xixin Zhao, Yaohua Zhou, Jiayi Zhang, Yuze Zhai, and Xiaocong Li. Opinion: Towards Unified Expressive Policy Optimization for Robust Robot Learning. *arXiv preprint arXiv:2511.10087*, 2025. URL <https://arxiv.org/abs/2511.10087>.
- [102] Kevin Huang, Rosario Scalise, Cleah Winston, Ayush Agrawal, Yunchu Zhang, Rohan Bajjal, Markus Grotz, Byron Boots, Benjamin Burchfiel, Masha Itkina, Paarth Shah, and Abhishek Gupta. Using Non-Expert Data to Robustify Imitation Learning via Offline Reinforcement Learning. *arXiv preprint arXiv:2510.19495*, 2025. URL <https://arxiv.org/abs/2510.19495>.
- [103] Shuaiyi Huang, Mara Levy, Anubhav Gupta, Daniel Ekpo, Ruijie Zheng, and Abhinav Shrivastava. TREND: Tri-teaching for Robust Preference-based Reinforcement Learning with Demonstrations. *arXiv preprint arXiv:2505.06079*, 2025. URL <https://arxiv.org/abs/2505.06079>.
- [104] Xingshuai Huang, Di Wu, and Benoit Boulet. Goal-Conditioned Data Augmentation for Offline Reinforcement Learning. *arXiv preprint arXiv:2412.20519*, 2024. URL <https://arxiv.org/abs/2412.20519>.
- [105] Zhiyu Huang, Xinshuo Weng, Maximilian Igl, Yuxiao Chen, Yulong Cao, Boris Ivanovic, Marco Pavone, and Chen Lv. Gen-Drive: Enhancing Diffusion Generative Driving Policies with Reward Modeling and Reinforcement Learning Fine-tuning. *arXiv preprint arXiv:2410.05582*, 2024. URL <https://arxiv.org/abs/2410.05582>.
- [106] Zilin Huang, Zihao Sheng, Zhengyang Wan, Yansong Qu, Junwei You, Sicong Jiang, and Sikai Chen. DriveVLM-RL: Neuroscience-Inspired Reinforcement Learning with Vision-Language Models for Safe and Deployable Autonomous Driving. *arXiv preprint arXiv:2603.18315*, 2026. URL <https://arxiv.org/abs/2603.18315>.
- [107] Dongjie Huo, Haoyun Liu, Guoqing Liu, Dekang Qi, Zhiming Sun, Maoguo Gao, Jianxin He, Yandan Yang, Xinyuan Chang, Feng Xiong, Xing Wei, Zhiheng Ma, and Mu Xu. ABot-Claw: A Foundation for Persistent, Cooperative, and Self-Evolving Robotic Agents. *arXiv preprint arXiv:2604.10096*, 2026. URL <https://arxiv.org/abs/2604.10096>.
- [108] Marcel Hussing, Jorge A. Mendez, Anisha Singrodia, Cassandra Kent, and Eric Eaton. Robotic Manipulation Datasets for Offline Compositional Reinforcement Learning. *arXiv preprint arXiv:2307.07091*, 2023. URL <https://arxiv.org/abs/2307.07091>.
- [109] Minyoung Hwang, Luca Weihs, Chanwoo Park, Kimin Lee, Aniruddha Kembhavi, and Kiana Ehsani. Promptable Behaviors: Personalizing Multi-Objective Rewards from Human Preferences. *arXiv preprint arXiv:2312.09337*, 2023. URL <https://arxiv.org/abs/2312.09337>.
- [110] Minyoung Hwang, Alexandra Forsey-Smerek, Nathaniel Dennler, and Andreea Bobu. Masked IRL: LLM-Guided Reward Disambiguation from Demonstrations and Language. *arXiv preprint arXiv:2511.14565*, 2025. URL <https://arxiv.org/abs/2511.14565>.
- [111] Yash Jangir, Yidi Zhang, Pang-Chi Lo, Kashu Yamazaki, Chenyu Zhang, Kuan-Hsun Tu, Tsung-Wei Ke, Lei Ke, Yonatan Bisk, and Katerina Fragkiadaki. RobotArena \$\infty\$: Scalable Robot Benchmarking via Real-to-Sim Translation. *arXiv preprint arXiv:2510.23571*, 2025. URL <https://arxiv.org/abs/2510.23571>.
- [112] Snehal Jauhri, Vignesh Prasad, and Georgia Chalvatzaki. Whole-Body Mobile Manipulation using Offline Reinforcement Learning on Sub-optimal Controllers. *arXiv preprint arXiv:2604.12509*, 2026. URL <https://arxiv.org/abs/2604.12509>.
- [113] Ke Jiang, Wen Jiang, Yao Li, and Xiaoyang Tan. Beyond Non-Expert Demonstrations: Outcome-Driven Action Constraint for Offline Reinforcement Learning. *arXiv preprint arXiv:2504.01719*, 2025. URL <https://arxiv.org/abs/2504.01719>.
- [114] Wonsuhk Jung, Utkarsh A. Mishra, Nadun Ranawaka Arachchige, Yongxin Chen, Danfei Xu, and Shreyas Kousik. Joint Model-based Model-free Diffusion for Planning with

- Constraints. *arXiv preprint arXiv:2509.08775*, 2025. URL <https://arxiv.org/abs/2509.08775>.
- [115] Swaminathan S K and Aritra Hazra. SPAARS: Safer RL Policy Alignment through Abstract Exploration and Refined Exploitation of Action Space. *arXiv preprint arXiv:2603.09378*, 2026. URL <https://arxiv.org/abs/2603.09378>.
- [116] Yuki Kadokawa, Jonas Frey, Takahiro Miki, Takamitsu Matsubara, and Marco Hutter. DAPPER: Discriminability-Aware Policy-to-Policy Preference-Based Reinforcement Learning for Query-Efficient Robot Skill Acquisition. *arXiv preprint arXiv:2505.06357*, 2025. URL <https://arxiv.org/abs/2505.06357>.
- [117] Elis Karcini, Faisal Mehrban, Quang Nguyen, Mac Schwager, Arash Ajoudani, Cesar Cadena, Jan Peters, Marco Hutter, and Haitham Bou-Ammar. Robots Need More than VLA and World Models. *arXiv preprint arXiv:2606.06556*, 2026. URL <https://arxiv.org/abs/2606.06556>.
- [118] Timo Kaufmann, Paul Weng, Viktor Bengs, and Eyke Hullermeier. A Survey of Reinforcement Learning from Human Feedback. *arXiv preprint arXiv:2312.14925*, 2023. URL <https://arxiv.org/abs/2312.14925>.
- [119] Rituraj Kaushik, Karol Arndt, and Ville Kyrki. SafeAPT: Safe Simulation-to-Real Robot Learning using Diverse Policies Learned in Simulation. *arXiv preprint arXiv:2201.13248*, 2022. URL <https://arxiv.org/abs/2201.13248>.
- [120] Richard Kelley. Behavior Trees Enable Structured Programming of Language Model Agents. *arXiv preprint arXiv:2404.07439*, 2024. URL <https://arxiv.org/abs/2404.07439>.
- [121] R. Khorrambakht, Joaquim Ortiz-Haro, Joseph Amigo, Omar Mostafa, Daniel Dugas, Franziska Meier, and Ludovic Righetti. WorldPlanner: Monte Carlo Tree Search and MPC with Action-Conditioned Visual World Models. *arXiv preprint arXiv:2511.03077*, 2025. URL <https://arxiv.org/abs/2511.03077>.
- [122] Changyeon Kim, Minho Heo, Doohyun Lee, Jinwoo Shin, Honglak Lee, Joseph J. Lim, and Kimin Lee. Subtask-Aware Visual Reward Learning from Segmented Demonstrations. *arXiv preprint arXiv:2502.20630*, 2025. URL <https://arxiv.org/abs/2502.20630>.
- [123] Seungsu Kim, Jinyoung Choi, Seungmin Baek, and Jean-Michel Renders. ProgVLA: Progress-Aware Robot Manipulation Skill Learning. *arXiv preprint arXiv:2605.28231*, 2026. URL <https://arxiv.org/abs/2605.28231>.
- [124] Suzie Kim. Reinforcement Learning from Implicit Neural Feedback for Human-Aligned Robot Control. *arXiv preprint arXiv:2512.00050*, 2025. URL <https://arxiv.org/abs/2512.00050>.
- [125] Suzie Kim, Hye-Bin Shin, and Seong-Whan Lee. Aligning Humans and Robots via Reinforcement Learning from Implicit Human Feedback. *arXiv preprint arXiv:2507.13171*, 2025. URL <https://arxiv.org/abs/2507.13171>.
- [126] Prajwal Koirala and Cody Fleming. Solving Offline Reinforcement Learning with Decision Tree Regression. *arXiv preprint arXiv:2401.11630*, 2024. URL <https://arxiv.org/abs/2401.11630>.
- [127] Jan Kompatscher, Danqing Shi, Giovanna Varni, Tino Weinkauff, and Antti Oulasvirta. Interactive Groupwise Comparison for Reinforcement Learning from Human Feedback. *arXiv preprint arXiv:2507.04340*, 2025. URL <https://arxiv.org/abs/2507.04340>.
- [128] Ksenia Konyushkova, Konrad Zolna, Yusuf Aytar, Alexander Novikov, Scott Reed, Serkan Cabi, and Nando de Freitas. Semi-supervised reward learning for offline reinforcement learning. *arXiv preprint arXiv:2012.06899*, 2020. URL <https://arxiv.org/abs/2012.06899>.
- [129] Pierre Krack, Tobias Julg, Wolfram Burgard, and Florian Walter. Rewarding DINO: Predicting Dense Rewards with Vision Foundation Models. *arXiv preprint arXiv:2603.16978*, 2026. URL <https://arxiv.org/abs/2603.16978>.
- [130] Sachit Kuhar, Shuo Cheng, Shivang Chopra, Matthew Bronars, and Danfei Xu. Learning to Discern: Imitating Heterogeneous Human Demonstrations with Preference and Representation Learning. *arXiv preprint arXiv:2310.14196*, 2023. URL <https://arxiv.org/abs/2310.14196>.
- [131] Ashish Kumar and Ilya Kuzovkin. Offline Robot Reinforcement Learning with Uncertainty-Guided Human Expert Sampling. *arXiv preprint arXiv:2212.08232*, 2022. URL <https://arxiv.org/abs/2212.08232>.
- [132] Aviral Kumar, Anikait Singh, Frederik Ebert, Mitsuhiro Nakamoto, Yanlai Yang, Chelsea Finn, and Sergey Levine. Pre-Training for Robots: Offline RL Enables Learning New Tasks from a Handful of Trials. *arXiv preprint arXiv:2210.05178*, 2022. URL <https://arxiv.org/abs/2210.05178>.
- [133] Edoardo Lamon, Fabio Fusaro, Elena De Momi, and Arash Ajoudani. A Unified Architecture for Dynamic Role Allocation and Collaborative Task Planning in Mixed Human-Robot Teams. *arXiv preprint arXiv:2301.08038*, 2023. URL <https://arxiv.org/abs/2301.08038>.
- [134] Alex X. Lee, Coline Devin, Jost Tobias Springenberg, Yuxiang Zhou, Thomas Lampe, Abbas Abdolmaleki, and Konstantinos Bousmalis. How

- to Spend Your Robot Time: Bridging Kickstarting and Offline Reinforcement Learning for Vision-based Robotic Manipulation. *arXiv preprint arXiv:2205.03353*, 2022. URL <https://arxiv.org/abs/2205.03353>.
- [135] Chris Lee, Flora Salim, Benjamin Tag, and Francisco Cruz. Generating Natural and Expressive Robot Gestures through Iterative Reinforcement Learning with Human Feedback using LLMs. *arXiv preprint arXiv:2606.18747*, 2026. URL <https://arxiv.org/abs/2606.18747>.
- [136] Kimin Lee, Laura Smith, and Pieter Abbeel. PEBBLE: Feedback-Efficient Interactive Reinforcement Learning via Relabeling Experience and Unsupervised Pre-training. *arXiv preprint arXiv:2106.05091*, 2021. URL <https://arxiv.org/abs/2106.05091>.
- [137] Seunghyun Lee, Younggyo Seo, Kimin Lee, Pieter Abbeel, and Jinwoo Shin. Offline-to-Online Reinforcement Learning via Balanced Replay and Pessimistic Q-Ensemble. *arXiv preprint arXiv:2107.00591*, 2021. URL <https://arxiv.org/abs/2107.00591>.
- [138] Tony Lee, Andrew Wagenmaker, Karl Pertsch, Percy Liang, Sergey Levine, and Chelsea Finn. RoboReward: General-Purpose Vision-Language Reward Models for Robotics. *arXiv preprint arXiv:2601.00675*, 2026. URL <https://arxiv.org/abs/2601.00675>.
- [139] Jacob Levy, Tyler Westenbroek, Kevin Huang, Fernando Palafox, Patrick Yin, Shayegan Omidshafiei, Dong-Ki Kim, Abhishek Gupta, and David Fridovich-Keil. Simulation Distillation: Pretraining World Models in Simulation for Rapid Real-World Adaptation. *arXiv preprint arXiv:2603.15759*, 2026. URL <https://arxiv.org/abs/2603.15759>.
- [140] Chao Li. Accelerating Self-Imitation Learning from Demonstrations via Policy Constraints and Q-Ensemble. *arXiv preprint arXiv:2212.03562*, 2022. URL <https://arxiv.org/abs/2212.03562>.
- [141] Chenliang Li, Siliang Zeng, Zeyi Liao, Jiaxiang Li, Dongyeop Kang, Alfredo Garcia, and Mingyi Hong. Learning Reward and Policy Jointly from Demonstration and Preference Improves Alignment. *arXiv preprint arXiv:2406.06874*, 2024. URL <https://arxiv.org/abs/2406.06874>.
- [142] Huanyu Li, Kun Lei, Sheng Zang, Kaizhe Hu, Yongyuan Liang, Bo An, Xiaoli Li, and Huazhe Xu. Failure-Aware RL: Reliable Offline-to-Online Reinforcement Learning with Self-Recovery for Real-World Manipulation. *arXiv preprint arXiv:2601.07821*, 2026. URL <https://arxiv.org/abs/2601.07821>.
- [143] Jin Li, Xianyuan Zhan, Zixu Xiao, and Guyue Zhou. Efficient Robotic Manipulation Through Offline-to-Online Reinforcement Learning and Goal-Aware State Information. *arXiv preprint arXiv:2110.10905*, 2021. URL <https://arxiv.org/abs/2110.10905>.
- [144] Jinghan Li, Zhicheng Sun, and Yadong Mu. Closed-Loop Long-Horizon Robotic Planning via Equilibrium Sequence Modeling. *arXiv preprint arXiv:2410.01440*, 2024. URL <https://arxiv.org/abs/2410.01440>.
- [145] Jinning Li, Xinyi Liu, Banghua Zhu, Jiantao Jiao, Masayoshi Tomizuka, Chen Tang, and Wei Zhan. Guided Online Distillation: Promoting Safe Reinforcement Learning by Offline Demonstration. *arXiv preprint arXiv:2309.09408*, 2023. URL <https://arxiv.org/abs/2309.09408>.
- [146] Shangzhe Li, Zhiao Huang, and Hao Su. Coupled Distributional Random Expert Distillation for World Model Online Imitation Learning. *arXiv preprint arXiv:2505.02228*, 2025. URL <https://arxiv.org/abs/2505.02228>.
- [147] Yunfei Li, Xiao Ma, Jiafeng Xu, Yu Cui, Zhongren Cui, Zhigang Han, Liqun Huang, Tao Kong, Yuxiao Liu, Hao Niu, Wanli Peng, Jingchao Qiao, Zeyu Ren, Haixin Shi, Zhi Su, Jiawen Tian, Yuyang Xiao, Shenyu Zhang, Liwei Zheng, Hang Li, et al. GR-RL: Going Dexterous and Precise for Long-Horizon Robotic Manipulation. *arXiv preprint arXiv:2512.01801*, 2025. URL <https://arxiv.org/abs/2512.01801>.
- [148] Yuxuan Li, Yicheng Gao, Ning Yang, and Stephen Xia. TW-CRL: Time-Weighted Contrastive Reward Learning for Efficient Inverse Reinforcement Learning. *arXiv preprint arXiv:2504.05585*, 2025. URL <https://arxiv.org/abs/2504.05585>.
- [149] Anthony Liang, Yigit Korkmaz, Jiahui Zhang, Minyoung Hwang, Abrar Anwar, Sidhant Kaushik, Aditya Shah, Alex S. Huang, Luke Zettlemoyer, Dieter Fox, Yu Xiang, Anqi Li, Andreea Bobu, Abhishek Gupta, Stephen Tu, Erdem Biyik, and Jesse Zhang. Robometer: Scaling General-Purpose Robotic Reward Models via Trajectory Comparisons. *arXiv preprint arXiv:2603.02115*, 2026. URL <https://arxiv.org/abs/2603.02115>.
- [150] Xinran Liang, Katherine Shu, Kimin Lee, and Pieter Abbeel. Reward Uncertainty for Exploration in Preference-based Reinforcement Learning. *arXiv preprint arXiv:2205.12401*, 2022. URL <https://arxiv.org/abs/2205.12401>.
- [151] Shuhao Liao, Xuxin Lv, Yuhong Cao, Jeric Lew, Wenjun Wu, and Guillaume Sartoretti. HELM: Human-Preferred Exploration with Language Models. *arXiv preprint arXiv:2503.07006*, 2025. URL <https://arxiv.org/abs/2503.07006>.

- [152] Haohong Lin, Wenhao Ding, Zuxin Liu, Yaru Niu, Jiacheng Zhu, Yuming Niu, and Ding Zhao. Safety-aware Causal Representation for Trustworthy Offline Reinforcement Learning in Autonomous Driving. *arXiv preprint arXiv:2311.10747*, 2023. URL <https://arxiv.org/abs/2311.10747>.
- [153] Haohong Lin, Radu Corcodel, and Ding Zhao. Generalize by Touching: Tactile Ensemble Skill Transfer for Robotic Furniture Assembly. *arXiv preprint arXiv:2404.17684*, 2024. URL <https://arxiv.org/abs/2404.17684>.
- [154] Huanrong Liu, Chunlin Tian, Tongyu Jia, Tailai Zhou, Qin Liu, Yu Gao, Yutong Ban, Yun Gu, Guy Rosman, Xin Ma, and Qingbiao Li. SutureFormer: Learning Surgical Trajectories via Goal-conditioned Offline RL in Pixel Space. *arXiv preprint arXiv:2603.26720*, 2026. URL <https://arxiv.org/abs/2603.26720>.
- [155] Jinyi Liu, Yifu Yuan, Jianye Hao, Fei Ni, Lingzhi Fu, Yibin Chen, and Yan Zheng. Enhancing Robotic Manipulation with AI Feedback from Multimodal Large Language Models. *arXiv preprint arXiv:2402.14245*, 2024. URL <https://arxiv.org/abs/2402.14245>.
- [156] Pengsen Liu, Maosen Zeng, Nan Tang, Kaiyuan Li, Jing-Cheng Pang, Yunan Liu, and Yang Yu. VL-GOR: Visual-Language Knowledge Guided Offline Reinforcement Learning for Generalizable Agents. *arXiv preprint arXiv:2603.22892*, 2026. URL <https://arxiv.org/abs/2603.22892>.
- [157] Runze Liu, Yali Du, Fengshuo Bai, Jiafei Lyu, and Xiu Li. PEARL: Zero-shot Cross-task Preference Alignment and Robust Reward Learning for Robotic Manipulation. *arXiv preprint arXiv:2306.03615*, 2023. URL <https://arxiv.org/abs/2306.03615>.
- [158] Runze Liu, Chenjia Bai, Jiafei Lyu, Shengjie Sun, Yali Du, and Xiu Li. VLP: Vision-Language Preference Learning for Embodied Manipulation. *arXiv preprint arXiv:2502.11918*, 2025. URL <https://arxiv.org/abs/2502.11918>.
- [159] Tenglong Liu, Yang Li, Yixing Lan, Hao Gao, Wei Pan, and Xin Xu. Adaptive Advantage-Guided Policy Regularization for Offline Reinforcement Learning. *arXiv preprint arXiv:2405.19909*, 2024. URL <https://arxiv.org/abs/2405.19909>.
- [160] Xiaoming Liu, Borong Zhang, Qingbiao Li, and Steven Morad. 120 Minutes and a Laptop: Minimalist Image-goal Navigation via Unsupervised Exploration and Offline RL. *arXiv preprint arXiv:2603.26441*, 2026. URL <https://arxiv.org/abs/2603.26441>.
- [161] Yuan Liu, Haoran Li, Shuai Tian, Yuxing Qin, Yuhui Chen, Yupeng Zheng, Yongzhen Huang, and Dongbin Zhao. Towards Long-Lived Robots: Continual Learning VLA Models via Reinforcement Fine-Tuning. *arXiv preprint arXiv:2602.10503*, 2026. URL <https://arxiv.org/abs/2602.10503>.
- [162] Yuyang Liu, Chuan Wen, Yihang Hu, Dinesh Jayaraman, and Yang Gao. TimeRewarder: Learning Dense Reward from Passive Videos via Frame-wise Temporal Distance. *arXiv preprint arXiv:2509.26627*, 2025. URL <https://arxiv.org/abs/2509.26627>.
- [163] Zeyi Liu, Guangyao Liu, Yinuo Qu, Yuquan Xue, Bofang Jia, Chunhua Yang, Weihua Gui, Keke Huang, and Ziwei Wang. Preference-Calibrated Human-in-the-Loop Reinforcement Learning for Robotic Manipulation. *arXiv preprint arXiv:2606.03949*, 2026. URL <https://arxiv.org/abs/2606.03949>.
- [164] Ziang Liu, Junjie Xu, Xingjiao Wu, Jing Yang, and Liang He. Multi-Type Preference Learning: Empowering Preference-Based Reinforcement Learning with Equal Preferences. *arXiv preprint arXiv:2409.07268*, 2024. URL <https://arxiv.org/abs/2409.07268>.
- [165] Guanxing Lu, Wenkai Guo, Chubin Zhang, Yuheng Zhou, Haonan Jiang, Zifeng Gao, Yansong Tang, and Ziwei Wang. VLA-RL: Towards Masterful and General Robotic Manipulation with Scalable Reinforcement Learning. *arXiv preprint arXiv:2505.18719*, 2025. URL <https://arxiv.org/abs/2505.18719>.
- [166] Jianlan Luo, Perry Dong, Jeffrey Wu, Aviral Kumar, Xinyang Geng, and Sergey Levine. Action-Quantized Offline Reinforcement Learning for Robotic Skill Learning. *arXiv preprint arXiv:2310.11731*, 2023. URL <https://arxiv.org/abs/2310.11731>.
- [167] Jianlan Luo, Charles Xu, Jeffrey Wu, and Sergey Levine. Precise and Dexterous Robotic Manipulation via Human-in-the-Loop Reinforcement Learning. *arXiv preprint arXiv:2410.21845*, 2024. URL <https://arxiv.org/abs/2410.21845>.
- [168] Tung M. Luu, Donghoon Lee, Younghwan Lee, and Chang D. Yoo. Policy Learning from Large Vision-Language Model Feedback without Reward Modeling. *arXiv preprint arXiv:2507.23391*, 2025. URL <https://arxiv.org/abs/2507.23391>.
- [169] Tung M. Luu, Hwanhee Kim, Younghwan Lee, and Chang D. Yoo. Video-Based Optimal Transport for Feedback-Efficient Offline Preference-Based Reinforcement Learning. *arXiv preprint arXiv:2606.16856*, 2026. URL <https://arxiv.org/abs/2606.16856>.
- [170] Tung Minh Luu, Younghwan Lee, Donghoon Lee, Sunho Kim, Min Jun Kim, and Chang D. Yoo. Enhancing Rating-Based Reinforcement

- Learning to Effectively Leverage Feedback from Large Vision-Language Models. *arXiv preprint arXiv:2506.12822*, 2025. URL <https://arxiv.org/abs/2506.12822>.
- [171] Rachel Ma, Jingyi Qu, Andreea Bobu, and Dylan Hadfield-Menell. Goal Inference from Open-Ended Dialog. *arXiv preprint arXiv:2410.13957*, 2024. URL <https://arxiv.org/abs/2410.13957>.
- [172] Yecheng Jason Ma, William Liang, Guanzhi Wang, De-An Huang, Osbert Bastani, Dinesh Jayaraman, Yuke Zhu, Linxi Fan, and Anima Anandkumar. Eureka: Human-Level Reward Design via Coding Large Language Models. *arXiv preprint arXiv:2310.12931*, 2023. URL <https://arxiv.org/abs/2310.12931>.
- [173] Yunchang Ma, Tenglong Liu, Yixing Lan, Xin Yin, Changxin Zhang, Xinglong Zhang, and Xin Xu. Diffusion Policies with Value-Conditional Optimization for Offline Reinforcement Learning. *arXiv preprint arXiv:2511.08922*, 2025. URL <https://arxiv.org/abs/2511.08922>.
- [174] Mael Macuglia, Paul Friedrich, and Giorgia Ramponi. Fine-tuning Behavioral Cloning Policies with Preference-Based Reinforcement Learning. *arXiv preprint arXiv:2509.26605*, 2025. URL <https://arxiv.org/abs/2509.26605>.
- [175] Harsh Maheshkeka, Zhixian Xie, Zhaoran Wang, and Wanxin Jin. Language-Model-Assisted Bi-Level Programming for Reward Learning from Internet Videos. *arXiv preprint arXiv:2410.09286*, 2024. URL <https://arxiv.org/abs/2410.09286>.
- [176] Ajay Mandlekar, Danfei Xu, Josiah Wong, Soroush Nasiriany, Chen Wang, Rohun Kulkarni, Li Fei-Fei, Silvio Savarese, Yuke Zhu, and Roberto Martin-Martin. What Matters in Learning from Offline Human Demonstrations for Robot Manipulation. *arXiv preprint arXiv:2108.03298*, 2021. URL <https://arxiv.org/abs/2108.03298>.
- [177] Sabariswaran Mani, Sreyas Venkataraman, Abhranil Chandra, Adyan Rizvi, Yash Sirvi, Soumojit Bhattacharya, and Aritra Hazra. DiffClone: Enhanced Behaviour Cloning in Robotics with Diffusion-Driven Policy Learning. *arXiv preprint arXiv:2401.09243*, 2024. URL <https://arxiv.org/abs/2401.09243>.
- [178] Daniel Marta, Simon Holk, Miguel Vasco, Jens Lundell, Timon Homberger, Finn Busch, Olov Andersson, Danica Kragic, and Iolanda Leite. FLoRA: Sample-Efficient Preference-based RL via Low-Rank Style Adaptation of Reward Functions. *arXiv preprint arXiv:2504.10002*, 2025. URL <https://arxiv.org/abs/2504.10002>.
- [179] Connor Mattson, Anurag Aribandi, and Daniel S. Brown. Representation Alignment from Human Feedback for Cross-Embodiment Reward Learning from Mixed-Quality Demonstrations. *arXiv preprint arXiv:2408.05610*, 2024. URL <https://arxiv.org/abs/2408.05610>.
- [180] Shaunak A. Mehta and Dylan P. Losey. Unified Learning from Demonstrations, Corrections, and Preferences during Physical Human-Robot Interaction. *arXiv preprint arXiv:2207.03395*, 2022. URL <https://arxiv.org/abs/2207.03395>.
- [181] Zhiting Mei, Tenny Yin, Ola Shorinwa, Apurva Badithela, Zhonghe Zheng, Joseph Bruno, Madison Bland, Lihan Zha, Asher Hancock, Jaime Fernandez Fisac, Philip Dames, and Anirudha Majumdar. Video Generation Models in Robotics – Applications, Research Challenges, Future Directions. *arXiv preprint arXiv:2601.07823*, 2026. URL <https://arxiv.org/abs/2601.07823>.
- [182] Zhen Meng, Kan Chen, Xiangmin Xu, Erwin Jose Lopez Pulgarin, Emma Li, Philip G. Zhao, and David Flynn. Preference-Driven Active 3D Scene Representation for Robotic Inspection in Nuclear Decommissioning. *arXiv preprint arXiv:2504.02161*, 2025. URL <https://arxiv.org/abs/2504.02161>.
- [183] Colin Merk, Ismail Geles, Jiaxu Xing, Angel Romero, Giorgia Ramponi, and Davide Scaramuzza. Learning Acrobatic Flight from Preferences. *arXiv preprint arXiv:2508.18817*, 2025. URL <https://arxiv.org/abs/2508.18817>.
- [184] Chenlin Ming, Zitong Wang, Boxuan Zhang, Zhanxiang Cao, Xiaoming Duan, and Jianping He. Stochastic Trajectory Optimization for Robotic Skill Acquisition From a Suboptimal Demonstration. *arXiv preprint arXiv:2408.03131*, 2024. URL <https://arxiv.org/abs/2408.03131>.
- [185] Yunyang Mo, Jian Li, Qiwei Wu, Yihang Kang, and Renjing Xu. OHP-RL: Online Human Preference as Guidance in Reinforcement Learning for Robot Manipulation. *arXiv preprint arXiv:2605.15971*, 2026. URL <https://arxiv.org/abs/2605.15971>.
- [186] Ni Mu, Yao Luan, Yiqin Yang, Bo Xu, and Qingshan Jia. S-EPOA: Overcoming the Indistinguishability of Segments with Skill-Driven Preference-Based Reinforcement Learning. *arXiv preprint arXiv:2408.12130*, 2024. URL <https://arxiv.org/abs/2408.12130>.
- [187] Mohamed Nabail, Leo Kaixuan Cheng, Jingmin Wang, and Nicholas Rhinehart. UBP2: Uncertainty-Balanced Preference Planning for Efficient Preference-based Reinforcement Learning. *arXiv preprint arXiv:2606.19328*, 2026. URL <https://arxiv.org/abs/2606.19328>.
- [188] Jordan Abi Nader, David Lee, Nathaniel Dennler, and Andreea Bobu. QuickLAP: Quick Language-

- Action Preference Learning for Semi-Autonomous Agents. *arXiv preprint arXiv:2511.17855*, 2025. URL <https://arxiv.org/abs/2511.17855>.
- [189] Mitsuhiro Nakamoto, Oier Mees, Aviral Kumar, and Sergey Levine. Steering Your Generalists: Improving Robotic Foundation Models via Value Guidance. *arXiv preprint arXiv:2410.13816*, 2024. URL <https://arxiv.org/abs/2410.13816>.
- [190] Austin Narcomey, Nathan Tsoi, Ruta Desai, and Marynel Vazquez. Learning Human Preferences Over Robot Behavior as Soft Planning Constraints. *arXiv preprint arXiv:2403.19795*, 2024. URL <https://arxiv.org/abs/2403.19795>.
- [191] Wen Zheng Terence Ng, Jianda Chen, Yuan Xu, and Tianwei Zhang. Latent Embedding Adaptation for Human Preference Alignment in Diffusion Planners. *arXiv preprint arXiv:2503.18347*, 2025. URL <https://arxiv.org/abs/2503.18347>.
- [192] Thanh Nguyen, Tung Luu, Tri Ton, Sungwoong Kim, and Chang D. Yoo. Uncertainty-Aware Rank-One MIMO Q Network Framework for Accelerated Offline Reinforcement Learning. *arXiv preprint arXiv:2602.19917*, 2026. URL <https://arxiv.org/abs/2602.19917>.
- [193] Hexian Ni, Tao Lu, Haoyuan Hu, Yinghao Cai, and Shuo Wang. SENIOR: Efficient Query Selection and Preference-Guided Exploration in Preference-based Reinforcement Learning. *arXiv preprint arXiv:2506.14648*, 2025. URL <https://arxiv.org/abs/2506.14648>.
- [194] Soichiro Nishimori, Xin-Qiang Cai, Johannes Ackermann, and Masashi Sugiyama. Offline Reinforcement Learning with Domain-Unlabeled Data. *arXiv preprint arXiv:2404.07465*, 2024. URL <https://arxiv.org/abs/2404.07465>.
- [195] Chihiro Noguchi and Takaki Yamamoto. Pseudo-Expert Regularized Offline RL for End-to-End Autonomous Driving in Photorealistic Closed-Loop Environments. *arXiv preprint arXiv:2512.18662*, 2025. URL <https://arxiv.org/abs/2512.18662>.
- [196] Yumi Omori, Zixuan Dong, and Keith Ross. Should We Ever Prefer Decision Transformer for Offline Reinforcement Learning? *arXiv preprint arXiv:2507.10174*, 2025. URL <https://arxiv.org/abs/2507.10174>.
- [197] Motoki Omura, Yusuke Mukuta, Kazuki Ota, Takayuki Osa, and Tatsuya Harada. Offline Reinforcement Learning with Wasserstein Regularization via Optimal Transport Maps. *arXiv preprint arXiv:2507.10843*, 2025. URL <https://arxiv.org/abs/2507.10843>.
- [198] Malayandi Palan, Nicholas C. Landolfi, Gleb Shevchuk, and Dorsa Sadigh. Learning Reward Functions by Integrating Human Demonstrations and Preferences. *arXiv preprint arXiv:1906.08928*, 2019. URL <https://arxiv.org/abs/1906.08928>.
- [199] Minting Pan, Yitao Zheng, Jiajian Li, Yunbo Wang, and Xiaokang Yang. Video-Enhanced Offline Reinforcement Learning: A Model-Based Approach. *arXiv preprint arXiv:2505.06482*, 2025. URL <https://arxiv.org/abs/2505.06482>.
- [200] Jongjin Park, Younggyo Seo, Jinwoo Shin, Honglak Lee, Pieter Abbeel, and Kimin Lee. SURF: Semi-supervised Reward Learning with Data Augmentation for Feedback-efficient Preference-based Reinforcement Learning. *arXiv preprint arXiv:2203.10050*, 2022. URL <https://arxiv.org/abs/2203.10050>.
- [201] Napoleon Paxton. The Theorems of Dr. David Blackwell and Their Contributions to Artificial Intelligence. *arXiv preprint arXiv:2604.06621*, 2026. URL <https://arxiv.org/abs/2604.06621>.
- [202] Andi Peng, Belinda Z. Li, Ilia Sucholutsky, Nishanth Kumar, Julie A. Shah, Jacob Andreas, and Andreea Bobu. Adaptive Language-Guided Abstraction from Contrastive Explanations. *arXiv preprint arXiv:2409.08212*, 2024. URL <https://arxiv.org/abs/2409.08212>.
- [203] Baorui Peng, Wenyao Zhang, Liang Xu, Zekun Qi, Jiazhao Zhang, Hongsi Liu, Wenjun Zeng, and Xin Jin. ReWorld: Multi-Dimensional Reward Modeling for Embodied World Models. *arXiv preprint arXiv:2601.12428*, 2026. URL <https://arxiv.org/abs/2601.12428>.
- [204] Sriyash Poddar, Yanming Wan, Hamish Ivison, Abhishek Gupta, and Natasha Jaques. Personalizing Reinforcement Learning from Human Feedback with Variational Preference Learning. *arXiv preprint arXiv:2408.10075*, 2024. URL <https://arxiv.org/abs/2408.10075>.
- [205] Xinyao Qin, Junjie Lu, Kaixin Wang, Chuheng Zhang, Sinjae Kang, Kimin Lee, Min Xu, Bin Liang, Jun Yang, and Li Zhao. Beyond Monotonic Progress: Retry-Supervised Value Learning for Robot Imitation. *arXiv preprint arXiv:2606.24633*, 2026. URL <https://arxiv.org/abs/2606.24633>.
- [206] Rafael Rafailov, Tianhe Yu, Aravind Rajeswaran, and Chelsea Finn. Offline Reinforcement Learning from Images with Latent Space Models. *arXiv preprint arXiv:2012.11547*, 2020. URL <https://arxiv.org/abs/2012.11547>.
- [207] Rafael Rafailov, Kyle Hatch, Victor Kolev, John D. Martin, Mariano Phielipp, and Chelsea Finn. MOTO: Offline Pre-training to Online Fine-tuning for Model-based Robot Learning. *arXiv preprint arXiv:2401.03306*, 2024. URL <https://arxiv.org/abs/2401.03306>.
- [208] Rafael Rafailov, Kyle Hatch, Anikait Singh, Laura

- Smith, Aviral Kumar, Ilya Kostrikov, Philippe Hansen-Estruch, Victor Kolev, Philip Ball, Jiajun Wu, Chelsea Finn, and Sergey Levine. D5RL: Diverse Datasets for Data-Driven Deep Reinforcement Learning. *arXiv preprint arXiv:2408.08441*, 2024. URL <https://arxiv.org/abs/2408.08441>.
- [209] Rathnam Vidushika Rasanji, Jin Wei-Kocsis, Jiansong Zhang, Dongming Gan, Ragu Athinarayanan, and Paul Asunda. Toward Deployable Multi-Robot Collaboration via a Symbolically-Guided Decision Transformer. *arXiv preprint arXiv:2508.13877*, 2025. URL <https://arxiv.org/abs/2508.13877>.
- [210] Allen Z. Ren, Anushri Dixit, Alexandra Bodrova, Sumeet Singh, Stephen Tu, Noah Brown, Peng Xu, Leila Takayama, Fei Xia, Jake Varley, Zhenjia Xu, Dorsa Sadigh, Andy Zeng, and Anirudha Majumdar. Robots That Ask For Help: Uncertainty Alignment for Large Language Model Planners. *arXiv preprint arXiv:2307.01928*, 2023. URL <https://arxiv.org/abs/2307.01928>.
- [211] Luke Rowe, Roger Girgis, Anthony Gosselin, Bruno Carrez, Florian Golemo, Felix Heide, Liam Paull, and Christopher Pal. CtRL-Sim: Reactive and Controllable Driving Agents with Offline Reinforcement Learning. *arXiv preprint arXiv:2403.19918*, 2024. URL <https://arxiv.org/abs/2403.19918>.
- [212] Achkan Salehi and Stephane Doncieux. Integrating LLMs and Decision Transformers for Language Grounded Generative Quality-Diversity. *arXiv preprint arXiv:2308.13278*, 2023. URL <https://arxiv.org/abs/2308.13278>.
- [213] Julia Santaniello, Matthew Russell, Benson Jiang, Donatello Sassaroli, Robert Jacob, and Jivko Sinapov. Towards Reinforcement Learning from Neural Feedback: Mapping fNIRS Signals to Agent Performance. *arXiv preprint arXiv:2511.12844*, 2025. URL <https://arxiv.org/abs/2511.12844>.
- [214] Thomas Schmied, Fabian Paischer, Vihang Patil, Markus Hofmarcher, Razvan Pascanu, and Sepp Hochreiter. Retrieval-Augmented Decision Transformer: External Memory for In-context RL. *arXiv preprint arXiv:2410.07071*, 2024. URL <https://arxiv.org/abs/2410.07071>.
- [215] Gershon Seneviratne, Jianyu An, Sahire Ellahy, Kasun Weerakoon, Mohamed Bashir Elnoor, Jonathan Deepak Kannan, Amogha Thalihalla Sunil, and Dinesh Manocha. HALO: Human Preference Aligned Offline Reward Learning for Robot Navigation. *arXiv preprint arXiv:2508.01539*, 2025. URL <https://arxiv.org/abs/2508.01539>.
- [216] Gershon Seneviratne, Jianyu An, Vaibhav Shende, Sahire Ellahy, Yaxita Amin, Kondapi Manasanjani, Samarth Chopra, Jonathan Deepak Kannan, and Dinesh Manocha. CHOP: Counterfactual Human Preference Labels Improve Obstacle Avoidance in Visuomotor Navigation Policies. *arXiv preprint arXiv:2603.02004*, 2026. URL <https://arxiv.org/abs/2603.02004>.
- [217] Pierre Sermanet, Kelvin Xu, and Sergey Levine. Unsupervised Perceptual Rewards for Imitation Learning. *arXiv preprint arXiv:1612.06699*, 2016. URL <https://arxiv.org/abs/1612.06699>.
- [218] Pierre Sermanet, Anirudha Majumdar, Alex Irpan, Dmitry Kalashnikov, and Vikas Sindhwani. Generating Robot Constitutions & Benchmarks for Semantic Safety. *arXiv preprint arXiv:2503.08663*, 2025. URL <https://arxiv.org/abs/2503.08663>.
- [219] Tharun Sethuraman, Subham Agrawal, Nils Dengler, Jorge de Heuvel, Teena Hassan, and Maren Bennewitz. Interpreting Context-Aware Human Preferences for Multi-Objective Robot Navigation. *arXiv preprint arXiv:2603.17510*, 2026. URL <https://arxiv.org/abs/2603.17510>.
- [220] Keshav Shankar, Dan Ding, and Wei Gao. Low-Burden LLM-Based Preference Learning: Personalizing Assistive Robots from Natural Language Feedback for Users with Paralysis. *arXiv preprint arXiv:2604.01463*, 2026. URL <https://arxiv.org/abs/2604.01463>.
- [221] Anukriti Singh, Amisha Bhaskar, Peihong Yu, Souradip Chakraborty, Ruthwik Dasyam, Amrit Bedi, and Pratap Tokekar. VARP: Reinforcement Learning from Vision-Language Model Feedback with Agent Regularized Preferences. *arXiv preprint arXiv:2503.13817*, 2025. URL <https://arxiv.org/abs/2503.13817>.
- [222] Avi Singh, Albert Yu, Jonathan Yang, Jesse Zhang, Aviral Kumar, and Sergey Levine. COG: Connecting New Skills to Past Experience with Offline Reinforcement Learning. *arXiv preprint arXiv:2010.14500*, 2020. URL <https://arxiv.org/abs/2010.14500>.
- [223] Samarth Sinha, Ajay Mandlikar, and Animesh Garg. S4RL: Surprisingly Simple Self-Supervision for Offline Reinforcement Learning. *arXiv preprint arXiv:2103.06326*, 2021. URL <https://arxiv.org/abs/2103.06326>.
- [224] Meng Song, Xuezhi Wang, Tanay Biradar, Yao Qin, and Manmohan Chandraker. A Minimalist Prompt for Zero-Shot Policy Learning. *arXiv preprint arXiv:2405.06063*, 2024. URL <https://arxiv.org/abs/2405.06063>.
- [225] Run Su, Hao Fu, Shuai Zhou, and Yingao Fu. Integrating Offline Pre-Training with Online Fine-Tuning: A Reinforcement Learning Approach for Robot Social Navigation. *arXiv preprint arXiv:2510.00466*, 2025. URL <https://arxiv.org/abs/2510.00466>.

- [226] Zhian Su, Weijie Kong, Haonan Dong, and Huixu Dong. IG-RFT: An Interaction-Guided RL Framework for VLA Models in Long-Horizon Robotic Manipulation. *arXiv preprint arXiv:2602.20715*, 2026. URL <https://arxiv.org/abs/2602.20715>.
- [227] Haowen Sun, Han Wang, Chengzhong Ma, Shaolong Zhang, Jiawei Ye, Xingyu Chen, and Xuguang Lan. PRISM: Projection-based Reward Integration for Scene-Aware Real-to-Sim-to-Real Transfer with Few Demonstrations. *arXiv preprint arXiv:2504.20520*, 2025. URL <https://arxiv.org/abs/2504.20520>.
- [228] Yirong Sun and Yanjun Chen. PRISM: Preference Refinement via Implicit Scene Modeling for 3D Vision-Language Preference-Based Reinforcement Learning. *arXiv preprint arXiv:2503.10177*, 2025. URL <https://arxiv.org/abs/2503.10177>.
- [229] Aamodh Suresh, Angelique Taylor, Laurel D. Riek, and Sonia Martinez. Robot Navigation in Risky, Crowded Environments: Understanding Human Preferences. *arXiv preprint arXiv:2303.08284*, 2023. URL <https://arxiv.org/abs/2303.08284>.
- [230] Huajie Tan, Sixiang Chen, Yijie Xu, Zixiao Wang, Yuheng Ji, Cheng Chi, Yaoxu Lyu, Zhongxia Zhao, Xiansheng Chen, Peterson Co, Shaoxuan Xie, Guocai Yao, Pengwei Wang, Zhongyuan Wang, and Shanghang Zhang. Robo-Dopamine: General Process Reward Modeling for High-Precision Robotic Manipulation. *arXiv preprint arXiv:2512.23703*, 2025. URL <https://arxiv.org/abs/2512.23703>.
- [231] Jinzhou Tan, Gabriel Adineera, and Jinoh Kim. ProgAgent: A Continual RL Agent with Progress-Aware Rewards. *arXiv preprint arXiv:2603.07784*, 2026. URL <https://arxiv.org/abs/2603.07784>.
- [232] Arsh Tangri, Ondrej Biza, Dian Wang, David Klee, Owen Howell, and Robert Platt. Equivariant Offline Reinforcement Learning. *arXiv preprint arXiv:2406.13961*, 2024. URL <https://arxiv.org/abs/2406.13961>.
- [233] Aleksandar Taranovic, Onur Celik, Niklas Freymuth, Ge Li, Serge Thilges, Huy Le, Tai Hoang, Rania Rayyes, and Gerhard Neumann. PAWS: Preference Learning with Advantage-Weighted Segments. *arXiv preprint arXiv:2606.11982*, 2026. URL <https://arxiv.org/abs/2606.11982>.
- [234] Jakob Thumm, Christopher Agia, Marco Pavone, and Matthias Althoff. Text2Interaction: Establishing Safe and Preferable Human-Robot Interaction. *arXiv preprint arXiv:2408.06105*, 2024. URL <https://arxiv.org/abs/2408.06105>.
- [235] Ran Tian, Chenfeng Xu, Masayoshi Tomizuka, Jitendra Malik, and Andrea Bajcsy. What Matters to You? Towards Visual Representation Alignment for Robot Learning. *arXiv preprint arXiv:2310.07932*, 2023. URL <https://arxiv.org/abs/2310.07932>.
- [236] Ran Tian, Yilin Wu, Chenfeng Xu, Masayoshi Tomizuka, Jitendra Malik, and Andrea Bajcsy. Maximizing Alignment with Minimal Feedback: Efficiently Learning Rewards for Visuomotor Robot Policy Alignment. *arXiv preprint arXiv:2412.04835*, 2024. URL <https://arxiv.org/abs/2412.04835>.
- [237] Ran Tian, Yilin Wu, and Andrea Bajcsy. Position: Good Embodied Reward Models Need Bad Behavior Data. *arXiv preprint arXiv:2606.01036*, 2026. URL <https://arxiv.org/abs/2606.01036>.
- [238] Wentong Tian, Jiyuan Zhao, Tianliang Yao, Yuxiang Fan, Zhengyu Shi, Dong Liu, and Peng Qi. Vision-Language Procedural Reasoning for Context-Aware Reward Modeling of Robotic Endovascular Guidewire Navigation. *arXiv preprint arXiv:2606.30698*, 2026. URL <https://arxiv.org/abs/2606.30698>.
- [239] Marcel Torne, Anubha Mahajan, Abhijnya Bhat, and Chelsea Finn. Freeform Preference Learning for Robotic Manipulation. *arXiv preprint arXiv:2606.32027*, 2026. URL <https://arxiv.org/abs/2606.32027>.
- [240] Yi-Shiuan Tung, Bradley Hayes, and Alessandro Roncone. CRED: Counterfactual Reasoning and Environment Design for Active Preference Learning. *arXiv preprint arXiv:2507.05458*, 2025. URL <https://arxiv.org/abs/2507.05458>.
- [241] Yi-Shiuan Tung, Gyanig Kumar, Wei Jiang, Bradley Hayes, and Alessandro Roncone. CRED: Counterfactual Reasoning and Environment Design for Active Preference Learning. *arXiv preprint arXiv:2603.08531*, 2026. URL <https://arxiv.org/abs/2603.08531>.
- [242] Sreyas Venkataraman, Yufei Wang, Ziyu Wang, Navin Sriram Ravie, Zackory Erickson, and David Held. Real-World Offline Reinforcement Learning from Vision Language Model Feedback. *arXiv preprint arXiv:2411.05273*, 2024. URL <https://arxiv.org/abs/2411.05273>.
- [243] Mrinal Verghese and Christopher G. Atkeson. Human Preference Modeling Using Visual Motion Prediction Improves Robot Skill Learning from Egocentric Human Video. *arXiv preprint arXiv:2602.11393*, 2026. URL <https://arxiv.org/abs/2602.11393>.
- [244] Mudit Verma and Subbarao Kambhampati. A State Augmentation based approach to Reinforcement Learning from Human Preferences. *arXiv preprint arXiv:2302.08734*, 2023. URL <https://arxiv.org/abs/2302.08734>.

- [245] Mudit Verma, Siddhant Bhambri, and Subbarao Kambhampati. Exploiting Unlabeled Data for Feedback Efficient Human Preference based Reinforcement Learning. *arXiv preprint arXiv:2302.08738*, 2023. URL <https://arxiv.org/abs/2302.08738>.
- [246] Homer Walke, Kevin Black, Abraham Lee, Moo Jin Kim, Max Du, Chongyi Zheng, Tony Zhao, Philippe Hansen-Estruch, Quan Vuong, Andre He, Vivek Myers, Kuan Fang, Chelsea Finn, and Sergey Levine. BridgeData V2: A Dataset for Robot Learning at Scale. *arXiv preprint arXiv:2308.12952*, 2023. URL <https://arxiv.org/abs/2308.12952>.
- [247] Sebastian Wallkötter, Mohamed Chetouani, and Ginevra Castellano. SLOT-V: Supervised Learning of Observer Models for Legible Robot Motion Planning in Manipulation. *arXiv preprint arXiv:2210.01412*, 2022. URL <https://arxiv.org/abs/2210.01412>.
- [248] Huaxiaoyue Wang, Nathaniel Chin, Gonzalo Gonzalez-Pumariaga, Xiangwan Sun, Neha Sunkara, Maximus Adrian Pace, Jeannette Bohg, and Sanjiban Choudhury. APRICOT: Active Preference Learning and Constraint-Aware Task Planning with LLMs. *arXiv preprint arXiv:2410.19656*, 2024. URL <https://arxiv.org/abs/2410.19656>.
- [249] Pengqin Wang, Meixin Zhu, and Shaojie Shen. Environment Transformer and Policy Optimization for Model-Based Offline Reinforcement Learning. *arXiv preprint arXiv:2303.03811*, 2023. URL <https://arxiv.org/abs/2303.03811>.
- [250] Ruiqi Wang, Weizheng Wang, and Byung-Cheol Min. Feedback-efficient Active Preference Learning for Socially Aware Robot Navigation. *arXiv preprint arXiv:2201.00469*, 2022. URL <https://arxiv.org/abs/2201.00469>.
- [251] Ruiqi Wang, Dezhong Zhao, Dayoon Suh, Ziqin Yuan, Guohua Chen, and Byung-Cheol Min. Personalization in Human-Robot Interaction through Preference-based Action Representation Learning. *arXiv preprint arXiv:2409.13822*, 2024. URL <https://arxiv.org/abs/2409.13822>.
- [252] Ruiqi Wang, Dezhong Zhao, Ziqin Yuan, Ike Obi, and Byung-Cheol Min. PrefCLM: Enhancing Preference-based Reinforcement Learning with Crowdsourced Large Language Models. *arXiv preprint arXiv:2407.08213*, 2024. URL <https://arxiv.org/abs/2407.08213>.
- [253] Ruiqi Wang, Dezhong Zhao, Ziqin Yuan, Tianyu Shao, Guohua Chen, Dominic Kao, Sungeun Hong, and Byung-Cheol Min. PRIME: Preference-based Reinforcement Learning with Multimodal Feedback and Trajectory Synthesis from Foundation Models. *arXiv preprint arXiv:2509.15607*, 2025. URL <https://arxiv.org/abs/2509.15607>.
- [254] Ruixiang Wang, Qingming Liu, Yueci Deng, Guiliang Liu, Zhen Liu, and Kui Jia. EVA: Aligning Video World Models with Executable Robot Actions via Inverse Dynamics Rewards. *arXiv preprint arXiv:2603.17808*, 2026. URL <https://arxiv.org/abs/2603.17808>.
- [255] Siyin Wang, Zhaoye Fei, Qinyuan Cheng, Shiduo Zhang, Panpan Cai, Jinlan Fu, and Xipeng Qiu. World Modeling Makes a Better Planner: Dual Preference Optimization for Embodied Task Planning. *arXiv preprint arXiv:2503.10480*, 2025. URL <https://arxiv.org/abs/2503.10480>.
- [256] Weizheng Wang, Ruiqi Wang, Le Mao, and Byung-Cheol Min. NaviSTAR: Socially Aware Robot Navigation with Hybrid Spatio-Temporal Graph Transformer and Preference Learning. *arXiv preprint arXiv:2304.05979*, 2023. URL <https://arxiv.org/abs/2304.05979>.
- [257] Weizheng Wang, Chao Yu, Yu Wang, and Byung-Cheol Min. Human-Robot Cooperative Distribution Coupling for Hamiltonian-Constrained Social Navigation. *arXiv preprint arXiv:2409.13573*, 2024. URL <https://arxiv.org/abs/2409.13573>.
- [258] Xianghui Wang, Siwei Cheng, Shanze Wang, Xinming Zhang, Dan Zhang, and Wei Zhang. Learning from Demonstration with Failure Awareness for Safe Robot Navigation. *arXiv preprint arXiv:2604.23360*, 2026. URL <https://arxiv.org/abs/2604.23360>.
- [259] Xiaofei Wang, Kimin Lee, Kourosh Hakhamaneshi, Pieter Abbeel, and Michael Laskin. Skill Preferences: Learning to Extract and Execute Robotic Skills from Human Feedback. *arXiv preprint arXiv:2108.05382*, 2021. URL <https://arxiv.org/abs/2108.05382>.
- [260] Yi Wang, Xinchun Li, Pengwei Xie, Pu Yang, Buqing Nie, Yunuo Cai, Qinglin Zhang, Chendi Qu, Jeffrey Wu, Jianheng Song, Xinlin Ren, Jingshun Huang, Mingjie Pan, Siyuan Feng, Zhi Chen, and Jianlan Luo. Learning While Deploying: Fleet-Scale Reinforcement Learning for Generalist Robot Policies. *arXiv preprint arXiv:2605.00416*, 2026. URL <https://arxiv.org/abs/2605.00416>.
- [261] Zihan Wang, Jianwen Li, Li-Fan Wu, and Nina Mahmoudian. Deployable Vision-driven UAV River Navigation via Human-in-the-loop Preference Alignment. *arXiv preprint arXiv:2511.01083*, 2025. URL <https://arxiv.org/abs/2511.01083>.
- [262] Sarthak Wanjari. From Robotics to Sepsis Treatment: Offline RL via Geometric Pessimism. *arXiv preprint arXiv:2602.08655*, 2026. URL <https://arxiv.org/abs/2602.08655>.

- [263] Shin Watanabe, Geir Horn, Jim Trresen, and Kai Olav Ellefsen. Offline Skill Generalization via Task and Motion Planning. *arXiv preprint arXiv:2311.14328*, 2023. URL <https://arxiv.org/abs/2311.14328>.
- [264] Bohan Wu, Feng Xu, Zhanpeng He, Abhi Gupta, and Peter K. Allen. SQUIRL: Robust and Efficient Learning from Video Demonstration of Long-Horizon Robotic Manipulation Tasks. *arXiv preprint arXiv:2003.04956*, 2020. URL <https://arxiv.org/abs/2003.04956>.
- [265] Feiyang Wu, Zhaoyuan Gu, Hanran Wu, Anqi Wu, and Ye Zhao. Infer and Adapt: Bipedal Locomotion Reward Learning from Demonstrations via Inverse Reinforcement Learning. *arXiv preprint arXiv:2309.16074*, 2023. URL <https://arxiv.org/abs/2309.16074>.
- [266] Sihao Wu, Xiaonan Si, Chi Xing, Jianhong Wang, Gaojie Jin, Guangliang Cheng, Lijun Zhang, and Xiaowei Huang. Preference Alignment on Diffusion Model: A Comprehensive Survey for Image Generation and Editing. *arXiv preprint arXiv:2502.07829*, 2025. URL <https://arxiv.org/abs/2502.07829>.
- [267] Xiefeng Wu, Mingyu Hu, and Shu Zhang. Off-Policy Actor-Critic with Sigmoid-Bounded Entropy for Real-World Robot Learning. *arXiv preprint arXiv:2601.15761*, 2026. URL <https://arxiv.org/abs/2601.15761>.
- [268] Yanru Wu, Weiduo Yuan, Ang Qi, Vitor Guizilini, Jiageng Mao, and Yue Wang. Large Reward Models: Generalizable Online Robot Reward Generation with Vision-Language Models. *arXiv preprint arXiv:2603.16065*, 2026. URL <https://arxiv.org/abs/2603.16065>.
- [269] Theodor Wulff, Federico Tavella, Rahul Singh Maharjan, Manith Adikari, and Angelo Cangelosi. Grounding Hierarchical Vision-Language-Action Models Through Explicit Language-Action Alignment. *arXiv preprint arXiv:2604.05614*, 2026. URL <https://arxiv.org/abs/2604.05614>.
- [270] Guanwen Xie, Jingzehua Xu, Ziqi Zhang, Xiangwang Hou, Dongfang Ma, Shuai Zhang, Yong Ren, and Dusit Niyato. Is FISHER All You Need in The Multi-AUV Underwater Target Tracking Task? *arXiv preprint arXiv:2412.03959*, 2024. URL <https://arxiv.org/abs/2412.03959>.
- [271] Junhong Xu, Kai Yin, Jason M. Gregory, and Lantao Liu. Causal Inference for De-biasing Motion Estimation from Robotic Observational Data. *arXiv preprint arXiv:2210.08679*, 2022. URL <https://arxiv.org/abs/2210.08679>.
- [272] Linjie Xu, Zhengyao Jiang, Jinyu Wang, Lei Song, and Jiang Bian. Mildly Constrained Evaluation Policy for Offline Reinforcement Learning. *arXiv preprint arXiv:2306.03680*, 2023. URL <https://arxiv.org/abs/2306.03680>.
- [273] Tengye Xu, Yangting Sun, Ziju Shen, Guanqi Chen, Zhen Fu, Chen yizhou, Hua Chen, and Jia Pan. Beyond Pixels: Learning Invariant Rewards for Real-World Robotics From a Few Demonstrations. *arXiv preprint arXiv:2605.22123*, 2026. URL <https://arxiv.org/abs/2605.22123>.
- [274] Dongsheng Yang, Qianying Liu, Wataru Sato, Takashi Minato, Chaoran Liu, and Shin'ya Nishida. HAPI: A Model for Learning Robot Facial Expressions from Human Preferences. *arXiv preprint arXiv:2503.17046*, 2025. URL <https://arxiv.org/abs/2503.17046>.
- [275] Jingyun Yang, Max Sobol Mark, Brandon Vu, Archit Sharma, Jeannette Bohg, and Chelsea Finn. Robot Fine-Tuning Made Easy: Pre-Training Rewards and Policies for Autonomous Real-World Reinforcement Learning. *arXiv preprint arXiv:2310.15145*, 2023. URL <https://arxiv.org/abs/2310.15145>.
- [276] Pengzhi Yang, Xinyu Wang, Pengyu Jing, Kehan Wen, Yiduo Qu, Zhenhao Huang, Minghao Fu, Xin Liu, Yaheng Shen, and Fan Shi. RARM: Confidence-Gated Progress Reward Modeling for RL in Manipulation. *arXiv preprint arXiv:2606.22027*, 2026. URL <https://arxiv.org/abs/2606.22027>.
- [277] Yang Yang, Bingjie Chen, Zihan Wang, Yizhe Li, Guoping Pan, Yi Cheng, and Houde Liu. Stage-Transition Dense Reward Modeling for Reinforcement Learning. *arXiv preprint arXiv:2606.31377*, 2026. URL <https://arxiv.org/abs/2606.31377>.
- [278] Justin Yu, Andrew Goldberg, Kavish Kondap, Karim El-Refaï, Ethan Ransing, Qianzhong Chen, Mac Schwager, Fred Shentu, Philipp Wu, and Ken Goldberg. WARP-RM: A Warp-Augmented Relative Progress Reward Model for Data Curation. *arXiv preprint arXiv:2606.28320*, 2026. URL <https://arxiv.org/abs/2606.28320>.
- [279] Tianhe Yu, Aviral Kumar, Yevgen Chebotar, Karol Hausman, Chelsea Finn, and Sergey Levine. How to Leverage Unlabeled Data in Offline Reinforcement Learning. *arXiv preprint arXiv:2202.01741*, 2022. URL <https://arxiv.org/abs/2202.01741>.
- [280] Ziqin Yuan, Ruiqi Wang, Dezhong Zhao, Baijian Yang, and Byung-Cheol Min. PrefMoE: Robust Preference Modeling with Mixture-of-Experts Reward Learning. *arXiv preprint arXiv:2605.00384*, 2026. URL <https://arxiv.org/abs/2605.00384>.
- [281] Maryam Zare, Parham M. Kebria, and Abbas Khosravi. Leveraging Optimal Transport for Enhanced Offline Reinforcement Learning in Surgical Robotic Environments. *arXiv preprint*

- arXiv:2310.08841*, 2023. URL <https://arxiv.org/abs/2310.08841>.
- [282] Lihan Zha, Yuchen Cui, Li-Heng Lin, Minae Kwon, Montserrat Gonzalez Arenas, Andy Zeng, Fei Xia, and Dorsa Sadigh. Distilling and Retrieving Generalizable Knowledge for Robot Manipulation via Language Corrections. *arXiv preprint arXiv:2311.10678*, 2023. URL <https://arxiv.org/abs/2311.10678>.
- [283] Shaopeng Zhai, Qi Zhang, Tianyi Zhang, Fuxian Huang, Haoran Zhang, Ming Zhou, Shengzhe Zhang, Litao Liu, Sixu Lin, and Jiangmiao Pang. A Vision-Language-Action-Critic Model for Robotic Real-World Reinforcement Learning. *arXiv preprint arXiv:2509.15937*, 2025. URL <https://arxiv.org/abs/2509.15937>.
- [284] Huixin Zhan, Feng Tao, and Yongcan Cao. Human-guided Robot Behavior Learning: A GAN-assisted Preference-based Reinforcement Learning Approach. *arXiv preprint arXiv:2010.07467*, 2020. URL <https://arxiv.org/abs/2010.07467>.
- [285] Arthur Zhang, Carl Qi, Donne Su, Xiangyun Meng, Amy Zhang, and Joydeep Biswas. Foresight: Iterative Reasoning About Clues that Matter for Navigation. *arXiv preprint arXiv:2606.12550*, 2026. URL <https://arxiv.org/abs/2606.12550>.
- [286] Chaofan Zhang, Peng Hao, Xiaoge Cao, Xiaoshuai Hao, Shaowei Cui, and Shuo Wang. VTLA: Vision-Tactile-Language-Action Model with Preference Learning for Insertion Manipulation. *arXiv preprint arXiv:2505.09577*, 2025. URL <https://arxiv.org/abs/2505.09577>.
- [287] Hongyin Zhang, Shiyuan Zhang, Junxi Jin, Qixin Zeng, Yifan Qiao, Hongchao Lu, and Donglin Wang. Balancing Signal and Variance: Adaptive Offline RL Post-Training for VLA Flow Models. *arXiv preprint arXiv:2509.04063*, 2025. URL <https://arxiv.org/abs/2509.04063>.
- [288] Hongyin Zhang, Zifeng Zhuang, Han Zhao, Pengxiang Ding, Hongchao Lu, and Donglin Wang. ReinboT: Amplifying Robot Visual-Language Manipulation with Reinforcement Learning. *arXiv preprint arXiv:2505.07395*, 2025. URL <https://arxiv.org/abs/2505.07395>.
- [289] Jiahui Zhang, Yusen Luo, Abrar Anwar, Sumedh Anand Sontakke, Joseph J Lim, Jesse Thomason, Erdem Biyik, and Jesse Zhang. ReWiND: Language-Guided Rewards Teach Robot Policies without New Demonstrations. *arXiv preprint arXiv:2505.10911*, 2025. URL <https://arxiv.org/abs/2505.10911>.
- [290] Shuoqin Zhang, Yixin Xiong, Xiru Gao, Kai Liu, Ke Wang, Xichuan Zhou, and Zhe Hu. RoHIL: Robust Human-in-the-Loop Robotic Reinforcement Learning Against Illumination Variations. *arXiv preprint arXiv:2605.19924*, 2026. URL <https://arxiv.org/abs/2605.19924>.
- [291] Dezhong Zhao, Ruiqi Wang, Dayoon Suh, Taehyeon Kim, Ziqin Yuan, Byung-Cheol Min, and Guohua Chen. PrefMMT: Modeling Human Preferences in Preference-based Reinforcement Learning with Multimodal Transformers. *arXiv preprint arXiv:2409.13683*, 2024. URL <https://arxiv.org/abs/2409.13683>.
- [292] Kaiyan Zhao, Borong Zhang, Yiming Wang, Xingyu Liu, Xuetao Li, Yuyang Chen, and Xiaoguang Niu. E²DT: Efficient and Effective Decision Transformer with Experience-Aware Sampling for Robotic Manipulation. *arXiv preprint arXiv:2605.00159*, 2026. URL <https://arxiv.org/abs/2605.00159>.
- [293] Yi Zhao, Rinu Boney, Alexander Ilin, Juho Kannala, and Joni Pajarinen. Adaptive Behavior Cloning Regularization for Stable Offline-to-Online Reinforcement Learning. *arXiv preprint arXiv:2210.13846*, 2022. URL <https://arxiv.org/abs/2210.13846>.
- [294] Fangwei Zhong, Kui Wu, Hai Ci, Churan Wang, and Hao Chen. Empowering Embodied Visual Tracking with Visual Foundation Models and Offline RL. *arXiv preprint arXiv:2404.09857*, 2024. URL <https://arxiv.org/abs/2404.09857>.
- [295] Gaoyue Zhou, Liyiming Ke, Siddhartha Srinivasa, Abhinav Gupta, Aravind Rajeswaran, and Vikash Kumar. Real World Offline Reinforcement Learning with Realistic Data Source. *arXiv preprint arXiv:2210.06479*, 2022. URL <https://arxiv.org/abs/2210.06479>.
- [296] Shuai Zhou, Hao Fu, Haodong He, and Wei Liu. Robot Crowd Navigation in Dynamic Environment with Offline Reinforcement Learning. *arXiv preprint arXiv:2312.11032*, 2023. URL <https://arxiv.org/abs/2312.11032>.
- [297] Yixiang Zhu, Yonghao Chen, Zijie Yang, Yuesong Hu, and Xinyu Chen. DEFLECT: Temporal Counterfactual Preference Learning for Delay-Robust Asynchronous VLAs. *arXiv preprint arXiv:2605.19294*, 2026. URL <https://arxiv.org/abs/2605.19294>.
- [298] Christos Ziakas, Alessandra Russo, and Avishek Joey Bose. Aligning Flow Map Policies with Optimal Q-Guidance. *arXiv preprint arXiv:2605.12416*, 2026. URL <https://arxiv.org/abs/2605.12416>.
- [299] Konrad Zolna, Alexander Novikov, Ksenia Konyushkova, Caglar Gulcehre, Ziyu Wang, Yusuf Aytar, Misha Denil, Nando de Freitas, and Scott Reed. Offline Learning from Demonstrations and Unlabeled Experience.

- arXiv preprint arXiv:2011.13885*, 2020. URL <https://arxiv.org/abs/2011.13885>.
- [300] Dominik Zurek, Kamil Faber, Marcin Pietron, Pawe Gajewski, and Roberto Corizzo. TSN-Affinity: Similarity-Driven Parameter Reuse for Continual Offline Reinforcement Learning. *arXiv preprint arXiv:2604.25898*, 2026. URL <https://arxiv.org/abs/2604.25898>.